

Staggered extraction for exact matrix multiplication over every field

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Abstract

We prove that the arithmetic exponent of square matrix multiplication over every fixed field satisfies $\omega < 2.371054886006746$. This includes every positive characteristic.

1 Introduction

For a fixed field $\mathbb{F}$, the matrix multiplication exponent $\omega = \omega_{\mathbb{F}}$ is the infimum of the real numbers τ for which two $n \times n$ matrices over $\mathbb{F}$ can be multiplied using $O(n^{\tau+o(1)})$ scalar additions and multiplications. This exponent measures the asymptotic arithmetic cost of a central bilinear operation and of many algorithms built from it. The field is fixed throughout; constants in an arithmetic algorithm may depend on the field and on its chosen finite construction.

Strassen’s seven-product algorithm for 2×2 matrices gave the subcubic exponent $\log_2 7$ [12]. Subsequent advances used tensors to record bilinear operations: tensor rank counts the products of linear forms needed to realize an operation. Approximate bilinear identities and their conversion to exact algorithms [4], Schönhage’s asymptotic sum inequality [10], and Strassen’s laser method [13] established the main ingredients of this approach. The sum inequality converts a low-rank direct sum of matrix tensors into an exponent bound. The laser method constructs such direct sums by restricting powers of a tensor with low asymptotic rank. Coppersmith and Winograd combined these ideas with progression-free sets and introduced the tensor family used here [5].

Passing to a larger tensor power gives more freedom in choosing the intermediate pieces before extracting matrix tensors from them. Stothers analyzed the fourth power, in work subsequently published with Davie [11, 6]. Vassilevska Williams analyzed the eighth power, and Le Gall continued through the thirty-second power [14, 9]. Alman and Vassilevska Williams then reduced the extraction loss that arises when prescribed marginal distributions permit several joint distributions, obtaining $\omega < 2.3728596$ [2].

A further source of loss is requiring entire variable blocks to be disjoint when the final matrix tensors need only disjoint subsets of those blocks. Duan, Wu, and Zhou used asymmetric hashing to reduce this *combination loss* in the recursive analysis [7]. Vassilevska Williams, Xu, Xu, and Zhou allowed finer sharing of variables and extended recovery from deletions on all three sides to the restricted tensor products needed in the recursion [15]. Alman, Duan, Vassilevska Williams, Xu, Xu, and Zhou then introduced successive compatibility tests that treat all three sides differently [1]. These tests underlie the extraction studied here.

For a recent numerical comparison, Dupont et al. obtained $\omega < 2.371177$ by optimizing the eighth-power instance of this asymmetric framework, combining modern optimization with AlphaEvolve [8]. Version 3 of *More Asymmetry Yields Faster Matrix Multiplication* records that improvement and credits the new parameters to Dupont et al. [1, Table 1]. We prove the following bound over every fixed field.

Theorem 1.1. *Over every fixed field $\mathbb{F}$, the arithmetic exponent of square matrix multiplication satisfies*

$$\omega_{\mathbb{F}} < 2.371054886006746 < 2.371054887.$$

The construction gives exact arithmetic algorithms, including in positive characteristic. Its finite choices are independent of the input matrices. The theorem concerns asymptotic arithmetic cost; the counting arguments and parameter choices do not give bit-complexity or practical crossover estimates.

1.1 Proof overview

We use the Coppersmith–Winograd tensor with parameter 5, denoted CW_5 . Section 2 defines it and proves that the rank of its m th tensor power is at most 7^m times a polynomial in m , over the original field. A *strand of length ℓ* is a copy of $\text{CW}_5^{\otimes \ell}$. On each side the variables of CW_5 are indexed by $0, \dots, 6$, with weights $0, 1, 1, 1, 1, 1, 2$. A strand variable is a word of these indices, and a *shape* records the weight sums on the three sides. We start with strands of length 8 and split them successively into lengths 4, 2, and 1. At each depth we restrict by shape. A shape with a zero coordinate already gives a matrix multiplication tensor; positive shapes pass to the next split.

Combining capacities across depths. An extraction restricts a tensor to a direct sum of specified tensor products. To make the outputs disjoint, its three side-assignment tests each impose a bound on the guaranteed logarithmic number of outputs; we call these bounds its *directional capacities*. The guarantee uses their minimum. The constituent-stage analysis in [1, Theorem 6.4] already combines differently prescribed populations at one depth before taking this minimum. Our joint step also combines populations at different depths, provided their priority orders agree on the physical tensor sides. For their weighted capacity vectors $v_r \in \mathbb{R}^3$, the benefit of pooling across depths is expressed by

$$\sum_r \min_W v_{r,W} \leq \min_W \sum_r v_{r,W}.$$

Thus a shortage on one side at one depth can be offset by capacity from another depth.

Section 3 proves this joint extraction by following the actual order of the tests. On the first side, variables are grouped by their sequences of strand or half-strand weight sums; these groups are the *coarse blocks*. Each retained coarse block is assigned to a unique output. Compatibility with that assignment then determines which complete variables on the second and third sides can be retained. The result is a direct sum of the intended products with some variables deleted. Populations retain their ancestry and physical placement throughout, so each subsequent split receives its own prescribed distributions.

Recovering outputs with inherited restrictions. Some variables are missing because an earlier extraction imposed type conditions, meaning conditions on empirical distributions of variable statistics. Others are deleted to resolve competing assignments in the current extraction. Earlier recovery arguments already treat three-side deletions and inherited type conditions [15, Section 7], [1, Section 6.5]. Section 4 proves the form needed for our joint step. With N the large strand-count parameter, the extraction selects outputs in which every orbit of complete variables under the allowed position permutations loses at most C/N of its variables, uniformly over the orbits. Here C depends only on the fixed population data and tolerances. Independent symmetry shifts then cover the ideal tensor. Partitioning variables by their presence in those shifts reconstructs every coefficient exactly once, over the original field. Only a subexponential number of independent input copies is needed. The recovered products can then be subdivided and used in further extractions.

Staggering and the final rank comparison. A *lot* contains N length-8 strands. We stagger finitely many lots so that, at an interior time step, one lot is at each of the three splitting stages. Stage priorities balance the summed directional capacities. The work is partitioned among the six physical axis orders; one joint extraction acts on each part, and the six output counts multiply. Section 5 constructs this schedule, shown in Figure 1, and computes its finite boundary loss. Section 6 counts the common oriented dimensions of the terminal matrix tensors and combines these counts with the schedule to prove the total yield and rank comparison.

Write H_0 for the initial logarithmic yield, C_* for the contribution of the three staggered stages, and S_* for the terminal logarithmic matrix volume, all normalized per initial length-8 strand. The direct-sum rank inequality then gives

$$\omega_{\mathbb{F}} \leq \frac{3(8 \log 7 - H_0 - C_*)}{S_*}. \quad (1)$$

For each fixed number of lots, we first take the divisible strand count N to infinity and remove approximation losses; the number of lots tends to infinity last. Thus the boundary cost vanishes without a uniform assertion for a growing schedule. The explicit rational data in Section 7 and Appendix A certify $S_* > 0$ and place the ratio in (1) below the bound in Theorem 1.1. The complete parameter verifier and its rational-logarithm interval certificate accompany the source. The proof requires a single feasible choice of parameters, with no assertion that this choice is optimal.

2 Rank and the basic tensor

Fix an arbitrary field $\mathbb{F}$. All tensor identities and linear maps below are over $\mathbb{F}$, and all logarithms are natural. A tensor is viewed as a trilinear form in three disjoint sets of variables. Its rank $R(T)$ is the least number of products of three linear forms, one on each side, whose sum is T . A *restriction* substitutes linear forms separately on the three sides. Restrictions cannot increase rank. A direct sum uses disjoint variables, and a tensor product pairs the variable indices on each side; rank is subadditive under direct sum and submultiplicative under tensor product. These statements follow by applying the indicated operations to rank decompositions.

For positive integers a, b, c , write

$$\langle a, b, c \rangle = \sum_{i=1}^a \sum_{j=1}^b \sum_{k=1}^c x_{ij} y_{jk} z_{ki}.$$

This is the tensor for multiplying an $a \times b$ matrix by a $b \times c$ matrix. Its *volume* is abc . Pairing the three matrix indices gives

$$\langle a, b, c \rangle \otimes \langle a', b', c' \rangle \cong \langle aa', bb', cc' \rangle.$$

Cyclically permuting the three variable spaces replaces the parameters by (b, c, a) or (c, a, b) and preserves rank.

2.1 Exact rank and the arithmetic exponent

Put $R_d = R(\langle d, d, d \rangle)$ and

$$w = \inf_{d \geq 2} \frac{\log R_d}{\log d}, \quad d \text{ an integer.} \quad (2)$$

The flattening that sends a z coordinate to its coefficient bilinear form has rank d^2 : the d^2 forms $\sum_j x_{ij} y_{jk}$ are linearly independent. A rank-one tensor has flattening rank at most one. Thus

$d^2 \leq R_d \leq d^3$ and $2 \leq w \leq 3$. In particular,

$$R_d \geq d^w \quad (d \geq 1), \quad (3)$$

where $R_1 = 1$ handles the unit case.

Let ω denote the arithmetic exponent of square matrix multiplication over $\mathbb{F}$, counting scalar additions and multiplications. A fixed rank- R_d decomposition gives a bilinear matrix multiplication identity with R_d products. This identity remains valid on matrix blocks: every product has its x -linear factor before its y -linear factor, and comparison of coefficients proves the same identity for noncommuting blocks with central coefficients in $\mathbb{F}$. At size d^k the resulting arithmetic cost satisfies

$$C_k \leq R_d C_{k-1} + O_d(d^{2k}).$$

The implied constant may depend on the chosen decomposition. Since $R_d \geq d^2$, this gives $C_k = O_d(kR_d^k)$, including the boundary case $R_d = d^2$. Padding an arbitrary size to the next power of d therefore gives cost $O_d(n^{\log_d R_d} \log(n+1))$. The logarithmic factor is absorbed by n^η for every $\eta > 0$; taking the infimum in (2) proves

$$\omega \leq w. \quad (4)$$

We also need a uniform upper bound on exact square rank. Given $\varepsilon > 0$, choose one integer $d \geq 2$ with $\theta = \log_d R_d < w + \varepsilon$. For $h \geq 1$, tensor powers and padding give

$$R(\langle h, h, h \rangle) \leq R_d^{\lceil \log_d h \rceil} \leq C_\varepsilon h^{w+\varepsilon}, \quad C_\varepsilon = R_d. \quad (5)$$

This argument uses one approximating dimension; it does not require the infimum defining w to be attained.

We use the following identical-summand form of Schönhage's asymptotic sum inequality [10], with a proof from exact rank.

Lemma 2.1 (Direct sums of identical matrix tensors). *Let s, a, b, c be positive integers. If*

$$R\left(\bigoplus_{j=1}^s \langle a, b, c \rangle\right) \leq R_*,$$

then

$$s(abc)^{w/3} \leq R_*. \quad (6)$$

Proof. First consider s copies of $\langle d, d, d \rangle$. When $s = 1$, the assertion $sd^w \leq R_*$ is (3). When $d = 1$, the direct sum is $\sum_{j=1}^s x_j y_j z_j$, whose flattening has rank s , so the assertion also holds. We may therefore assume $s, d \geq 2$.

Fix $\varepsilon > 0$ and the constant in (5). For all sufficiently large integers k , set

$$h_k = \left\lfloor (s^k / C_\varepsilon)^{1/(w+\varepsilon)} \right\rfloor.$$

Then $h_k \geq 2$, $R(\langle h_k, h_k, h_k \rangle) \leq s^k$, and

$$\lim_{k \rightarrow \infty} \frac{\log h_k}{k} = \frac{\log s}{w + \varepsilon}.$$

The k th tensor power of the given direct sum is a direct sum of s^k copies of $\langle d^k, d^k, d^k \rangle$, of rank at most R_*^k . A rank- r decomposition of $\langle h_k, h_k, h_k \rangle$, with $r \leq s^k$, expresses it as a restriction of r

independent scalar products. Tensor this restriction with $\langle d^k, d^k, d^k \rangle$ and discard unused copies. It gives $\langle d^k h_k, d^k h_k, d^k h_k \rangle$ as a restriction of the available direct sum. Consequently

$$R_*^k \geq (d^k h_k)^w.$$

Taking logarithms, dividing by k , and then taking $k \rightarrow \infty$ at fixed ε yields

$$\log R_* \geq w \log d + \frac{w}{w + \varepsilon} \log s.$$

Letting $\varepsilon \downarrow 0$ proves $sd^w \leq R_*$.

For the rectangular case, let $U = \bigoplus_{j=1}^s \langle a, b, c \rangle$ and tensor U with its two cyclic rotations. Distributivity gives s^3 independent copies of

$$\langle a, b, c \rangle \otimes \langle b, c, a \rangle \otimes \langle c, a, b \rangle \cong \langle abc, abc, abc \rangle,$$

and the product has rank at most R_*^3 . The square case gives $s^3(abc)^w \leq R_*^3$, whose cube root is (6). The same proof covers $abc = 1$ through the unit square case. $\square$

The hypothesis in Lemma 2.1 specifies one common *oriented* triple (a, b, c) for all summands. In the construction below, exact history counts and common terminal type restrictions will provide these identical dimensions.

2.2 An integer coefficient construction

The relation between approximate bilinear identities and exact algorithms was studied by Bini [4]. Here a direct coefficient calculation gives an exact rank bound over the original field, including when that field is finite.

For an integer $q \geq 1$, define the symmetric Coppersmith–Winograd tensor [5]

$$\begin{aligned} \text{CW}_q = \sum_{i=1}^q & (x_0 y_i z_i + x_i y_0 z_i + x_i y_i z_0) \\ & + x_{q+1} y_0 z_0 + x_0 y_{q+1} z_0 + x_0 y_0 z_{q+1}. \end{aligned}$$

We use $q = 5$. The following exact-rank estimate makes the field independence and the polynomial overhead explicit.

Lemma 2.2 (Rank bound for powers of the basic tensor). *Over every field and for every integer $m \geq 0$,*

$$\text{R}(\text{CW}_q^{\otimes m}) \leq (q+2)^m \binom{3m+2}{2}. \quad (7)$$

Proof. We use the Laurent-polynomial identity of Coppersmith and Winograd [5, Section 7, equation (10)] over the integers, with a formal variable t . Set $X_1 = \sum_{i=1}^q x_i$, $Y_1 = \sum_{i=1}^q y_i$, and $Z_1 = \sum_{i=1}^q z_i$, and consider

$$\begin{aligned} P(t) = t^{-2} \sum_{i=1}^q & (x_0 + t x_i)(y_0 + t y_i)(z_0 + t z_i) \\ & - t^{-3}(x_0 + t^2 X_1)(y_0 + t^2 Y_1)(z_0 + t^2 Z_1) \\ & + (t^{-3} - q t^{-2})(x_0 + t^3 x_{q+1})(y_0 + t^3 y_{q+1})(z_0 + t^3 z_{q+1}). \end{aligned} \quad (8)$$

The only possible negative degrees are $-3, -2, -1$. Their coefficients are, respectively,

$$(-1+1)x_0y_0z_0, \quad (q-q)x_0y_0z_0, \quad (1-1)(X_1y_0z_0 + x_0Y_1z_0 + x_0y_0Z_1).$$

The degree-zero coefficient is precisely CW_q . Thus $P(t) = \text{CW}_q + tQ(t)$ for a polynomial Q with integer coefficients. In particular,

$$[t^0]P(t)^{\otimes m} = \text{CW}_q^{\otimes m}. \quad (9)$$

These coefficient identities remain valid after reduction to any field.

Regard (8) as $q+2$ products of three Laurent-polynomial linear forms, absorbing the scalar prefactor of each product into its x form. In one tensor position the possible degrees on the x side lie, respectively, in

$$\{-2, -1\}, \quad \{-3, -1\}, \quad \{-3, -2, 0, 1\}.$$

The last set includes both terms of $t^{-3} - qt^{-2}$ within a single linear form. On each of the other two sides the degrees lie between 0 and 3.

Expanding $P(t)^{\otimes m}$ therefore gives $(q+2)^m$ products, each of the form $A(t)B(t)C(t)$ with

$$A(t) = \sum_{a=-3m}^m A_a t^a, \quad B(t) = \sum_{b=0}^{3m} B_b t^b, \quad C(t) = \sum_{c=0}^{3m} C_c t^c.$$

Here each coefficient is one linear form in the corresponding tensor-product variables. The constant coefficient of this product is

$$\sum_{\substack{b,c \geq 0 \\ b+c \leq 3m}} A_{-b-c} B_b C_c,$$

a sum of at most $\binom{3m+2}{2}$ rank-one tensors. Summing over the $(q+2)^m$ choices and using (9) proves (7). For $m=0$, both sides equal 1 under the empty tensor product convention. Only coefficient addition and multiplication were used; no interpolation or field division is needed. $\square$

2.3 Strands and shapes

Give the label 0 weight 0, the labels $1, \dots, q$ weight 1, and the label $q+1$ weight 2. Every term of CW_q has total weight 2. A *strand of length ℓ* is a copy of $\text{CW}_q^{\otimes \ell}$; its variable indices are words of length ℓ . For a triple $g = (g_0, g_1, g_2)$ of nonnegative integers with $g_0 + g_1 + g_2 = 2\ell$, let $T_{\ell,g}$ be the restriction in which the weight sums on the three sides are g_0, g_1, g_2 . The coordinates of g refer to sides, whereas the weights 0, 1, 2 refer to individual labels. A shape is *positive* if all three coordinates are positive. When ℓ is even, splitting a strand into its two consecutive halves gives child shapes u and $g-u$, where $0 \leq u \leq g$ coordinatewise and $\sum_i u_i = \ell$.

A zero coordinate gives an especially simple terminal tensor. Suppose $g_0 = 0$. The x word is then forced to be the all-zero word. In each position a label on the y side uniquely determines the z label: 0 and $q+1$ are exchanged, and every label $i \in \{1, \dots, q\}$ is fixed. Thus the other two sides are matched bijectively, and $T_{\ell,g}$ is $\langle 1, 1, M \rangle$, where

$$M = [z^{g_1}](1 + qz + z^2)^\ell.$$

For $g_1 = 0$ or $g_2 = 0$ the corresponding orientations are $\langle M, 1, 1 \rangle$ or $\langle 1, M, 1 \rangle$. Restricting the matched words by corresponding type conditions on their two sides preserves this form, with M replaced by the number of allowed matched pairs; an empty matching gives the zero tensor. This observation will supply the terminal volume counts.

3 A joint extraction at several depths

The constituent-stage analysis of [1, Theorem 6.4] combines differently prescribed populations at a common depth. The extraction below also permits populations at different depths to share one hashing step, provided their priority orders agree on the physical tensor sides. Population data, strand lengths, and the constant sums defining their coarse supports may differ. The successive compatibility checks follow [1, Section 4.1], building on the asymmetric hashing and interface-tensor methods of [7, 15]. We specify the intended output first, then derive the three capacities that govern its extraction yield.

For a probability vector p on a finite alphabet, write $H(p) = -\sum_a p(a) \log p(a)$, with $0 \log 0 = 0$. A word has *type* p if its empirical distribution is exactly p ; a type window of width θ means coordinatewise distance at most θ . All alphabets, populations, and positive population masses in this section are fixed as N tends to infinity.

3.1 The data of a joint step

Index the populations by a finite set $\mathcal{R}$. Population r has $m_r = c_r N$ copies of a fixed strand tensor F_r with finite variable sets, where $c_r > 0$ is rational. On each side its variables are partitioned into coarse blocks indexed by $\{0, \dots, 16\}$. Every monomial of F_r has its triple of coarse indices in an ambient support Ω_r , whose elements $u = (u_X, u_Y, u_Z)$ satisfy

$$u_X + u_Y + u_Z = s_r. \quad (10)$$

Here X, Y, Z is one common ordering of the physical sides. In the product $\bigotimes_r F_r^{\otimes m_r}$, a coarse block is specified by a sequence of indices, one for every strand. A fine side variable is a complete basis variable of the product input; its coarse block records the coarse index at every strand position. The fixed strand tensor in a population is repeated identically at all its positions. Populations with different histories or physical placements are kept distinct.

Choose a rational law p_r on Ω_r that maximizes entropy among all laws on Ω_r with its three marginals. The support of p_r may be smaller than Ω_r . There are two kinds of population.

1. An *ordinary population* delivers the full tensor block at each selected triple u , without further output type tests. The blocks may themselves be tensor products of child strands. This includes extraction by whole-strand shape and the final split into length-one strands.
2. A *sharing population* consists of copies of $T_{\ell, g}$, with even ℓ and $\sum_W g_W = 2\ell$. Its coarse index u_W is the weight sum in the left half, so

$$\Omega_r = \{u : 0 \leq u \leq g, \sum_W u_W = \ell\}, \quad T_{\ell, g}|_u = T_{\ell/2, u} \otimes T_{\ell/2, g-u}.$$

Assume $p_r(u) = p_r(g - u)$. On each side W , a statistic of the full half-strand variable takes values in a fixed finite alphabet $\mathcal{A}_{r, W}$. A map ρ_W from this alphabet to weights determines the half weight sum. For every half-shape u with $p_r(u) > 0$, prescribe a rational law $\nu_{r, u, W}$ supported on $\rho_W^{-1}(u_W)$.

When a half-shape has a zero coordinate, the full variables on the other two sides match by complementation. Require that their statistics also match by a fixed bijection and that their prescribed laws are its pushforwards. A side having a unique possible variable has its corresponding point-mass prescription. These requirements apply as well when two coordinates vanish.

Ideal outputs and inherited masks. A *target triple* e is a triple of coarse sequences having exact joint type p_r in each population. Start with the tensor block at e . In every sharing population, impose a positive type window around $\nu_{r,u,W}$ separately in every class of positions of given (r, u, half) , on each side. The right-half prescription in the class indexed by u is $\nu_{r,g-u,W}$. Leave ordinary blocks unrestricted. The resulting tensor product is the *ideal output* Q_e . Target triples have the same class sizes, so their ideal outputs are isomorphic by position permutations preserving the physical sides; denote a canonical such tensor by Q_N . A fine side variable in a coarse block of e is *useful for* e if it satisfies all these output windows; ordinary populations contribute no further condition. Thus usefulness specifies the variables of the ideal tensor. Additional input or separation tests may delete some of them.

The child windows constrain the two halves separately. Inherited conditions can also constrain how their statistics are paired at parent positions. For a sharing population, suppress r temporarily and define the mixture law, omitting all zero-mass shapes as we do below:

$$D_W = \sum_{u \in \Omega_r} p(u) \nu_{u,W} \otimes \nu_{g-u,W}. \quad (11)$$

This is the expected pair law when, within each shape class, the left and right statistics have their prescribed laws and are independently arranged.

An input mask is always a deletion of individual side variables. For sharing populations we allow finitely many empirical tests on the ordered pair of half statistics, or a fixed function of that pair. Their centers must be the law $D_{r,W}$ in (11), or its corresponding pushforward, and their widths must be fixed positive numbers. Ordinary input masks must already follow from the exact target coarse types and the specified ideal blocks. No other inherited masks are assumed. All masks are imposed separately in the populations just defined. A mask on a larger received class can instead be enforced separately, with tighter widths and the same center, on any fixed subdivision; the resulting partwise restrictions imply the original one.

Choose output widths small enough that replacing each prescribed child law by any law in its window leaves the expected ordered-pair law strictly inside every inherited window. This condition has a uniform positive margin: for coordinatewise child errors at most θ , a product-law coordinate changes by at most 2θ , and all relevant collections are finite. The same margin will be required for the additional pair tests introduced in the proof.

Compatibility groups and capacities. The extraction will first separate targets by their X coarse blocks and impose the X output windows. It then separates Y variables, imposes their output windows, and finally separates Z variables. The later-side separation tests use only those individual shape types that matching forces from an earlier side. Define their designated shapes by

$$\mathcal{I}_Y = \{u \in \text{supp } p : u_Z = 0\}, \quad \mathcal{I}_Z = \{u \in \text{supp } p : u_X u_Y = 0\}. \quad (12)$$

A zero Z coordinate lets matching transfer the X -side output type condition to Y . For Z , a zero Y coordinate transfers the condition from X ; a zero X coordinate instead uses Y , after its output windows have been imposed. If both X and Y coordinates vanish, the Z variable is forced. For the remaining shapes with a given W coordinate, the pair law D_W determines only an aggregate statistic law after the designated contributions are removed. We impose no individual compatibility test on these shapes. The exact predicate and its tolerances are given in the incidence count below. For $W \in \{Y, Z\}$, the partition $\mathcal{G}_W$ consists of the individual groups $\{u\}$ for $u \in \mathcal{I}_W$, and the residual groups $\{u \in \text{supp } p \setminus \mathcal{I}_W : u_W = i\}$. Empty groups are omitted. A residual group is treated as

residual even when it contains just one shape. Put $p(G) = \sum_{u \in G} p(u)$ and define

$$J_W = \sum_{G \in \mathcal{G}_W} p(G) H \left(\frac{\sum_{u \in G} p(u) \nu_{u,W}}{p(G)} \right), \quad W \in \{Y, Z\}. \quad (13)$$

The entropy J_W measures the choices for one half-statistic word within these groups. Counting both halves gives the cost $2J_W$; comparison with the pair-word entropy $H(D_W)$ gives the saving that controls the number of compatible targets. Accordingly, define the sharing capacities by

$$C_{r,X} = H(p_{r,X}), \quad C_{r,Y} = H(D_{r,Y}) - 2J_{r,Y}, \quad C_{r,Z} = H(D_{r,Z}) - 2J_{r,Z}. \quad (14)$$

Only the law of half-shapes is assumed invariant under $u \mapsto g - u$; the two prescriptions $\nu_{u,W}$ and $\nu_{g-u,W}$ may differ.

For an ordinary population no compatibility restriction is needed, and the capacities are simply

$$C_{r,X} = H(p_{r,X}), \quad C_{r,Y} = H(p_{r,Y}), \quad C_{r,Z} = H(p_{r,Z}).$$

The counting arguments below give a common interpretation: each $C_{r,W}$ is the entropy saving in the bound on competitors at side W . The joint step adds these savings over populations before taking their minimum.

Theorem 3.1 (Heterogeneous joint step). *Let B_N be the restriction of $\bigotimes_r F_r^{\otimes m_r}$ by the inherited masks just specified, with the population data above. Set*

$$A_{\text{amb}} = \sum_{r \in \mathcal{R}} c_r H(p_r), \quad C_W = \sum_{r \in \mathcal{R}} c_r C_{r,W}, \quad E = \min\{C_X, C_Y, C_Z\}. \quad (15)$$

If $E > 0$, then for every $\varepsilon > 0$ one can choose positive output widths, as small as desired, such that for all sufficiently large N divisible by the fixed type denominators, a direct sum of $\exp(o(N))$ independent copies of B_N has a linear restriction to at least $\lfloor \exp(N(E - \varepsilon)) \rfloor$ independent copies of Q_N .

The choices of populations, rational laws, positive widths, and divisibility requirements precede the limit $N \rightarrow \infty$. In particular, the assertion does not require common strand lengths.

We first bound hash survival and assignment collisions. The sequential X, Y, Z tests then put surviving monomials in a direct sum of variable restrictions of the ideal outputs. Section 4 completes the proof by bounding the missing fraction on every full-variable orbit and restoring the ideal outputs from $\exp(o(N))$ input copies.

3.2 Entropy and a common modular hash

For a fixed alphabet $\mathcal{A}$, the number of words of type p is

$$\frac{m!}{\prod_a (mp(a))!} = \exp(mH(p) + O(\log(m+2))). \quad (16)$$

Indeed, comparing the sum $\sum_{j=1}^n \log j$ with its integral gives $\log(n!) = n \log n - n + O(\log(n+2))$. There are at most $(m+1)^{|\mathcal{A}|}$ possible types. Both bounds are uniform on each fixed alphabet, including types with zero coordinates.

One convenient way to meet the entropy hypothesis is to use positive weights a_i, b_j, c_k and set

$$p(i, j, k) = Z^{-1} a_i b_j c_k \quad ((i, j, k) \in \Omega_r).$$

For any competing law q with the same marginals, $-\sum q \log p = -\sum p \log p = H(p)$, because $\log p$ is a sum of three one-coordinate functions and a constant. Further,

$$\sum_{q(u)>0} q(u) \log \frac{p(u)}{q(u)} \leq \sum_{q(u)>0} (p(u) - q(u)) \leq 0,$$

by $\log t \leq t - 1$. Thus $H(q) \leq H(p)$. Positive rational weights give rational laws with this same property. When the marginals specify a unique law, maximality also holds, including boundary laws with zero probabilities.

First keep, on each side, just the coarse sequences having marginal type $p_{r,W}$ in every population. Call a triple satisfying these three filters and the support conditions *ambient*. The total number of target triples is

$$|\mathcal{T}_N| = \exp(N A_{\text{amb}} + O(\log N)). \quad (17)$$

For a fixed X coarse sequence, each possible ambient joint type q_r has the prescribed marginals. Its number of completions in population r is

$$\frac{\prod_i (m p_{r,X}(i))!}{\prod_u (m q_r(u))!} = \exp(m(H(q_r) - H(p_{r,X})) + O(\log N)).$$

Summing over the polynomially many types and using maximality yields the degree bound

$$\#\{\text{ambient triples at a fixed } X \text{ block}\} \leq \exp(N(A_{\text{amb}} - C_X) + O(\log N)). \quad (18)$$

Here and below the coarse counts ignore all fine masks, thereby bounding every interaction that could actually occur.

Fix a small slack $\delta > 0$. Since $C_X \leq A_{\text{amb}}$ and $E \leq C_X$, choose a power $M = 37^k$ with

$$\log M = N(A_{\text{amb}} - E + \delta) + O(1). \quad (19)$$

There is a set $U \subset \mathbb{Z}/M\mathbb{Z}$ with

$$\log |U| = \log M - O(\sqrt{\log M}) \quad (20)$$

and no nonconstant three-term arithmetic progression. Here is a version of Behrend's construction [3] that also rules out modular wraparound. Put $h = \lfloor \sqrt{\log M} \rfloor$, $b = \lfloor M^{1/h}/2 \rfloor$, and encode $d \in \{0, \dots, b-1\}^h$ as $n(d) = \sum_{j=0}^{h-1} d_j (2b)^j$. Choose a largest class with fixed $\sum_j d_j^2$. Every encoded integer is less than $(2b)^h/2 \leq M/2$, so a modular progression among them is an integer progression. In $n(d) + n(e) = 2n(f)$ all digit sums are at most $2b-2$, so there is no carry and $d+e=2f$. Equal squared norms now give $\|d-e\|^2 = 2\|d\|^2 + 2\|e\|^2 - \|d+e\|^2 = 0$. There are at most $1 + h(b-1)^2$ norm classes, whence $|U| \geq b^h/(1 + hb^2)$. The inequalities $b \geq M^{1/h}/4$ and $b \leq M^{1/h}$ imply (20).

Concatenate all $d = \sum_r m_r$ strand positions to form coarse words I, J, K . Let S have entry s_r at positions in population r . Every ambient triple satisfies $I + J + K = S$, even though the entries of S can vary. Choose independent uniform offsets a, b' and a uniform vector v over $\mathbb{Z}/M\mathbb{Z}$, and set

$$x(I) = a + \langle v, I \rangle, \quad y(J) = b' + \langle v, J \rangle, \quad z(K) = \frac{a + b' + \langle v, S - K \rangle}{2}. \quad (21)$$

Division by 2 is defined since M is odd. Keep only coarse blocks whose hashes lie in U . Since $x + y = 2z$, a surviving triple has $x = y = z$.

For a fixed ambient triple e , let S_e be its full hash-survival event. For every v and $t \in U$ there is exactly one offset pair giving $x = y = z = t$, so

$$\Pr(S_e) = h_M := \frac{|U|}{M^2}. \quad (22)$$

Conditional on S_e , (v, t) is uniform on $(\mathbb{Z}/M\mathbb{Z})^d \times U$. A second triple with differences $(\Delta I, \Delta J, \Delta K)$ has hashes

$$t + \langle v, \Delta I \rangle, \quad t + \langle v, \Delta J \rangle, \quad t - \frac{1}{2} \langle v, \Delta K \rangle.$$

If it shares X with e , survival is equivalent to $\langle v, \Delta J \rangle = 0$; if it shares Y or Z , it is equivalent to $\langle v, \Delta I \rangle = 0$. In the shared- Z case use $\Delta J = -\Delta I$. Distinct triples cannot share two blocks, by their common sum S . Thus the tested difference vector has a nonzero coordinate of absolute value at most 16, which is a unit modulo 37^k . Conditioning on the other coordinates of v shows, in all three cases, that

$$\Pr(S_{e'} \mid S_e) = \frac{1}{M} \quad \text{for distinct triples sharing a block.} \quad (23)$$

No further hash-dependent event is included in this conditioning.

3.3 The finite compatibility count

In a sharing population, prefilter side $W \in \{Y, Z\}$ by requiring its empirical ordered-pair law to be within width η of D_W . A fine side variable is *compatible* with a target triple if its statistic type is within width χ of the prescribed law in every designated shape class, separately on the two halves. On the left these are the positions with $u \in \mathcal{I}_W$, using $\nu_{u,W}$; on the right they are the positions with $g - u \in \mathcal{I}_W$, using $\nu_{g-u,W}$. No individual compatibility test is imposed on a residual group, including a residual group with one shape. The positions of each tested class are read from the target triple. Ordinary populations impose no compatibility test. A variable for the whole product is compatible exactly when it is compatible in every population.

Lemma 3.2 (Compatibility incidence count). *For fixed population data, there is a function $\xi(\eta, \chi) \rightarrow 0$ as $(\eta, \chi) \rightarrow (0, 0)$ such that any fine side variable passing the pair filters is compatible with at most*

$$\exp(N(A_{\text{amb}} - C_W + \xi(\eta, \chi)) + O(\log N)), \quad W \in \{Y, Z\}, \quad (24)$$

target triples containing its coarse block, before hashing. The estimate is uniform in the variable and the target arrangement.

Proof. Consider first one sharing population of size m , and fix the coarse W word $k = (k_1, \dots, k_m)$. If k does not have type p_W , no target contains it and the bound is immediate. Assume henceforth that it has this type, and let $\mathcal{C}$ be the set of shape words $(u_1, \dots, u_m)$ of type p with $u_{j,W} = k_j$. Such a word determines a complete coarse triple. If the given fine variable has pair-statistic word $z = ((a_j, b_j))_{j=1}^m$ and empirical law D' , let $\mathcal{B}$ be its orbit under

$$\Gamma_k = \prod_i \text{Sym}\{j : k_j = i\}.$$

This group acts transitively on $\mathcal{B}$ by definition and on $\mathcal{C}$ because all shape counts within every coarse class are fixed. Compatibility is unchanged by applying the same permutation to a candidate and a statistic word. Hence its incidence degrees $a = \#\{z \in \mathcal{B} : z \text{ compatible with } c\}$ and

$b = \#\{c \in \mathcal{C} : z \text{ compatible with } c\}$ do not depend on c and z , respectively. Counting incidences gives the exact finite identity

$$|\mathcal{C}|a = |\mathcal{B}|b. \quad (25)$$

Since the statistic determines the half weight,

$$\begin{aligned} |\mathcal{C}| &= \frac{\prod_i (mp_W(i))!}{\prod_u (mp(u))!}, & \log |\mathcal{C}| &= m(H(p) - H(p_W)) + O(\log(m+2)), \\ |\mathcal{B}| &= \frac{\prod_i (mp_W(i))!}{\prod_{a,b} (mD'(a,b))!}, & \log |\mathcal{B}| &= m(H(D') - H(p_W)) + O(\log(m+2)). \end{aligned}$$

To bound a , now fix a candidate $c \in \mathcal{C}$. This fixes the position set of every left and right compatibility group. On the left, compatibility specifies the law within each designated group to accuracy χ . The row sums of D' give the total count of each statistic among positions with each coarse value, since $\rho_W(a_j) = k_j$. Subtracting the designated-group counts leaves, in the remaining group with coarse value i , the law

$$\frac{\sum_{u \in G} p(u) \nu_{u,W}}{p(G)}$$

up to an error tending uniformly to zero with η and χ . More explicitly, each coordinate error is at most $(|\mathcal{A}_{r,W}|\eta + \chi)/p(G)$: summing a row of D' costs at most $|\mathcal{A}_{r,W}|\eta$, and the weighted errors of the subtracted designated laws total at most χ . Only fixed positive numbers $p(G)$ are divided by. A product of multinomial counts over these fixed position sets, and then a sum over polynomially many types, bounds the number of left-statistic assignments by $\exp(m(J_W + o_{\eta,\chi}(1)) + O(\log(m+2)))$.

For the right half, use the column sums of D' . Its coarse value is $g_W - k_j$, so those sums determine precisely the analogous aggregate counts. The right half-shape distribution is $p(g-u) = p(u)$, giving the same entropy J_W , with the actual right prescription $\nu_{g-u,W}$. Count the two halves independently; discarding the requirement that they reassemble the exact pair law D' can only enlarge the count. We obtain

$$a \leq \exp(m(2J_W + o_{\eta,\chi}(1)) + O(\log(m+2))).$$

There is no factor for choosing group locations: those locations were fixed with c . By (25) and uniform continuity of entropy on a finite simplex,

$$b \leq \exp(m(H(p) - H(D_W) + 2J_W + o_{\eta,\chi}(1)) + O(\log(m+2))).$$

For an ordinary population all candidates are compatible, and their number is $\exp(m(H(p) - H(p_W)) + O(\log(m+2)))$ directly. The candidate sets, permutation groups, and compatibility conditions factor over populations. Multiplying their bounds proves (24). $\square$

3.4 Sequential assignment by variable deletion

Choose $\eta, \chi > 0$ sufficiently small that the error in Lemma 3.2 is below $\delta/4$. Next choose output widths $\theta > 0$ smaller than χ and small enough for the positive margins required by all inherited and pair tests. Complementary prescribed laws ensure that usefulness of one side transfers to the compatibility window on a matching side. Every choice here is fixed before N grows.

First delete an X coarse block unless it belongs to exactly one ambient hash survivor and that survivor is a target triple. Call the retained triples *eligible*. Freeze this list for all subsequent tests. By (18), (19), and (23), a target triple e fails this test, conditional only on S_e , with probability at most $\exp(-\delta N + O(\log N))$. Each retained X block now has an unambiguous assignment. In that

block retain only the fine variables useful for its assigned triple, that is, those satisfying its output windows.

On Y , apply its pair filters and delete every fine variable having other than exactly one compatible eligible triple containing its coarse block. Assign it to that triple and impose usefulness for that assignment. Apply the same procedure to Z , using the designated classes $\mathcal{I}_Z$. Include all inherited masks as side-variable deletions; their application does not change the frozen eligible list or the definitions of the compatible lists.

We verify that these operations produce a direct sum, rather than merely a list of proposed components. Take an actual monomial surviving every deletion, and let e be its ambient coarse triple. Its X block survived, so e is that block's unique eligible triple and its X variable is useful for e . In every sharing half with $u_Z = 0$, the Y statistics are complementary to the X statistics. Thus the actual Y variable is compatible with e ; deterministic sides satisfy their point-mass tests as well. The Y uniqueness test therefore assigns this variable to e . Its usefulness is now known for the actual triple.

For Z , a half with $u_Y = 0$ transfers the required statistics from X . A half with $u_X = 0$ and $u_Y > 0$ transfers them from the already useful Y variable. With both $u_X = u_Y = 0$, the remaining variable is forced. These cases prove compatibility with e on every designated shape class in $\mathcal{I}_Z$. Its uniqueness test then assigns the actual Z variable to e as well. Ordinary populations add no compatibility conditions in either argument. All three variables of every retained monomial therefore have the same assignment.

Different assignments share no retained fine variable on any side. Within assignment e , usefulness restricts precisely to Q_e ; all other tests delete individual variables of Q_e . Consequently the surviving tensor is a direct sum of tensors Q_e^{hole} , each obtained from its ideal Q_e solely by variable deletion. This statement concerns complete variables of the population product; sharing a local coordinate factor between two complete variables does not identify those variables.

Here is the probability estimate needed to repair these holes. For a fixed e and a fixed variable v of Q_e passing all its deterministic inherited and pair masks, own-triple compatibility and usefulness hold. Let $R_{e,v}$ be the union of the X eligibility-failure event and, when v is on side Y or Z , the events that another compatible target triple sharing its coarse block survives the hash. Every random deletion of v is contained in $R_{e,v}$. Bound those competitors among *all* target triples before eligibility and other deletions. Lemma 3.2 and (23) then give, for some constant $c > 0$,

$$\Pr(R_{e,v} \mid S_e) \leq \exp(-cN), \quad (26)$$

uniformly in e and the passing variable, for all large N . Indeed, the exponent for side W is at most $N(E - C_W - \delta + \xi) + O(\log N)$ and $C_W \geq E$. This estimate is conditioned only on S_e , never on eligibility or on passing a hash-dependent test.

For reference in the recovery argument, let G_e act by independent position permutations in every ideal child class (r, u, half) , simultaneously on all three sides. In ordinary whole-strand blocks use their position permutations; for a factorized ordinary block one may also permute its child factors independently. These are automorphisms of Q_e . The full variable alphabets in its finitely many classes are fixed, so G_e has only polynomially many orbits of variables. Lemma 4.1 supplies the uniform deterministic mask bound on these orbits. The completion of Theorem 3.1 in Section 4 combines that bound with (26), selects exponentially many good assignments using (17) and (22), and applies Lemma 4.2 to reconstruct their ideal outputs.

4 Recovering the ideal outputs and composing extractions

The assignment procedure in Section 3 leaves a direct sum of variable restrictions of the intended outputs. We now select sufficiently many of these outputs and recover them exactly. Two different probability spaces occur: permutations of an ideal output control inherited masks, whereas the random hash controls assignment collisions. Keeping these spaces separate is essential when a received population already has type restrictions. Recovery from deletions on all three sides appears in [15, Section 7, especially Property 7.1 and Theorem 7.2]; its interface-tensor consequence is Corollary 4.2 there. The constituent-stage analysis in [1, Section 6.5] also accounts for inherited approximate type restrictions. Here we establish uniform loss on every full-variable orbit of each selected heterogeneous output and reconstruct its coefficients by an exact linear restriction.

4.1 Uniform control of inherited masks

Fix all population masses, rational laws, and positive tolerances before letting the divisible integer N grow. Classes of zero mass are omitted. For a target coarse triple e , write Q_e for its ideal child product, including its prescribed child-side type windows, but before inherited input masks and global pair-frequency masks. Keep distinct the population, its previous history, its exact parent split triple, and its left or right half. Each resulting child class c has $n_c = c_c N$ positions, with $c_c > 0$ fixed. At an ordinary extraction the corresponding whole-strand classes, or the unrestricted child classes of an ordinary split, are used instead.

Let G_e be the product of the symmetric groups on these classes. A permutation acts simultaneously on the three sides of a child tensor, and different child classes, including opposite halves, are permuted independently. This action preserves both the tensor factors and the child-side type windows, so it consists of coefficient-preserving automorphisms of Q_e . The action is on complete variables: a symbol records the entire child word, including its weight-one labels.

Lemma 4.1 (Uniform loss from inherited masks). *The total number of G_e -orbits of variables on the three sides is polynomial in N , uniformly over target triples e . Suppose every inherited or global mask is either implied by the exact coarse choices, or tests an empirical ordered pair of child statistics (or a fixed function of that pair) against its prescribed mixture law with a fixed positive tolerance. Choose the child tolerances sufficiently smaller than those mask tolerances, with the mixture laws consistent as in Theorem 3.1. There is then a constant C_0 such that, in every full-variable orbit on every side, the fraction failing any of these deterministic masks is at most C_0/N . The constant may depend on the fixed data and tolerances, but not on e , the orbit, or N .*

Proof. Let $\mathcal{A}_{c,W}$ be the finite alphabet of complete child variables on side W in class c . An orbit is specified by the multiplicity of every symbol in every class. Its number on side W is therefore at most

$$\prod_c (n_c + 1)^{|\mathcal{A}_{c,W}|} = O(N^{d_W})$$

for a fixed integer d_W . Child type windows merely select some of these orbits. The alphabets include all labels, even when a mask uses only a much coarser statistic.

Consider one mask with tolerance $\eta > 0$, on a population of size $m = cN$. Its tested law is an ordered-pair mixture. Write its exact parent-triple classes as a , with weights $w_a = n_a/m$, and let $q_{a,L}, q_{a,R}$ be the two empirical statistic laws in a fixed orbit. For the prescribed laws $\nu_{a,L}, \nu_{a,R}$, the child windows give coordinate errors at most τ . If a coarsening of a child statistic is needed, choose its original window smaller by the fixed alphabet size, so this assertion holds for the statistic

actually tested. The elementary inequality

$$|q_{a,L}(s)q_{a,R}(t) - \nu_{a,L}(s)\nu_{a,R}(t)| \leq 2\tau$$

shows that the mean mixture differs from $\sum_a w_a \nu_{a,L} \otimes \nu_{a,R}$ by at most 2τ in each coordinate. More generally, suppose the mask center differs from this prescribed mixture by coordinatewise error at most $\alpha \leq \eta/4$. Choosing $\tau \leq \eta/8$ then leaves a margin of at least $\eta/2$ inside the mask's acceptance interval. For the consistent rational prescriptions of Theorem 3.1, $\alpha = 0$. This margin holds also for orbits at the boundary of a child window.

Uniformly permute an element of the orbit. Within one parent-triple class of size n , the two half-statistic words are independently arranged. If the frequencies of s and t are a and b , their paired count $K_{s,t}$ satisfies, for $n \geq 2$,

$$\mathbb{E}K_{s,t} = nab, \quad \text{Var}(K_{s,t}) = \frac{n^2 a(1-a)b(1-b)}{n-1} \leq \frac{n}{8}. \quad (27)$$

Indeed, fix the positions of the na occurrences on the left. The indicators that the right symbol is t have mean b , variance $b(1-b)$, and covariance $-b(1-b)/(n-1)$ at distinct positions. Summing gives (27). This calculation is uniform over $a, b \in [0, 1]$, including deterministic symbols.

Different parent-triple classes use independent permutations. Consequently the variance of a pair count summed over a population is at most $m/8$. Chebyshev's inequality and the acceptance margin give

$$\Pr \left(\left| \frac{\sum_a K_{s,t}^{(a)}}{m} - \mathbb{E} \frac{\sum_a K_{s,t}^{(a)}}{m} \right| > \eta/2 \right) \leq \frac{1}{2c\eta^2 N}.$$

For a mask on a fixed function of the ordered pair, a coordinate of the pushed-forward law is a sum of at most B pair coordinates, where B is fixed. Choose $\tau \leq \eta/(8B)$ and apply the last bound to each pair coordinate with deviation threshold $\eta/(2B)$. Summing coordinate errors and taking a finite union bound proves the same $O(1/N)$ conclusion for that mask. All positive classes eventually have size at least two. There are only finitely many masks and statistic coordinates, so another union bound gives C_0/N . A uniform group element sends any fixed variable uniformly over its full orbit; thus this probability is exactly the rejected fraction of that orbit. We have not conditioned on passing a mask. Taking a minimum of the finitely many positive tolerance gaps and positive population masses makes C_0 uniform as asserted. A statistic that is itself an ordered pair is simply a symbol of a larger fixed alphabet. Masks implied by the exact coarse choices cause no loss. $\square$

4.2 Exact repair

We next give an algebraic recovery statement. A variable restriction of a tensor means that some basis variables are set to zero; all coefficients on the remaining variables retain their original values.

Lemma 4.2 (Exact repair by presence masks). *Let Q_N be a tensor over an arbitrary field, with at most $\exp(bN)$ nonzero monomials for a fixed $b > 0$. Let G_N be a finite group of coefficient-preserving permutations of its variables, acting separately on the three sides. Suppose a variable restriction H_N of Q_N omits at most C/N of each G_N -orbit of variables, for a fixed $C > 0$. For all sufficiently large N , a direct sum of*

$$R_N = \exp(O(N/\log N))$$

copies of H_N has Q_N as a linear restriction. Every coefficient is reconstructed exactly once. The bound is uniform over all H_N, Q_N, G_N satisfying these same constants.

Proof. We first cover all ideal monomials by shifted copies. Partitioning variables by their presence masks then converts this cover into an exact linear restriction. Put $\gamma = C/N$ and, for $N > 3C$, choose

$$L = \left\lceil \frac{(b+1)N}{\log(N/(3C))} \right\rceil = O(N/\log N).$$

Take L independent uniform elements $g_1, \dots, g_L$ of G_N . For a fixed ideal variable, its probability of absence from $g_i H_N$ is at most γ . For a fixed nonzero monomial xyz of Q_N , the union bound on its three variables makes its probability of absence at most 3γ . Independence of the L shifts, followed by a union bound over the support, bounds the probability of an uncovered monomial by

$$\exp(bN)(3\gamma)^L \leq e^{-N} < 1.$$

Fix a family covering every monomial. The group need only preserve Q_N ; it is allowed to move the holes of H_N .

For an ideal variable x , define its presence mask

$$A(x) = \{i \in \{1, \dots, L\} : x \text{ is retained in } g_i H_N\},$$

and define $B(y)$ and $C(z)$ similarly. Partition the three sides into the classes X_A, Y_B, Z_C of equal masks. There are at most 2^L classes on each side. For each triple of classes whose rectangle $X_A \times Y_B \times Z_C$ contains a nonzero monomial of Q_N , coverage implies $A \cap B \cap C \neq \emptyset$. Choose one index $i(A, B, C)$ in this intersection. All variables in these three classes are present in $g_{i(A, B, C)} H_N$. Since this is a variable restriction of Q_N , its restriction to these classes is the full ideal tensor on the rectangle, with every original coefficient.

Use a separate copy of H_N for each such rectangle, permute that copy by the chosen $g_{i(A, B, C)}$, and zero all other variables. In the direct sum of these rectangular tensors, map the copied variable (A, B, C, x) to the ideal variable x , and similarly on the other two sides. These are ordinary linear maps. Every nonzero monomial xyz belongs to exactly one triple of mask classes, so its coefficient is supplied once and only once. Identifying a variable used in several rectangles introduces no additional monomials: linear maps act on the sum of the existing tensor terms. There is no division or characteristic restriction. At most 2^{3L} copies are used, and padding with unused copies gives the asserted common bound R_N . $\square$

4.3 Completing the joint extraction

Completion of the proof of Theorem 3.1. Use the hash slack $\delta > 0$ from Section 3, with all entropy errors chosen smaller than a fixed fraction of this slack. For a target triple e , let S_e be its own hash-survival event; $\Pr(S_e) = h_M = |U|/M^2$. Let $P_e(v)$ denote the deterministic assertion that an ideal variable v of Q_e passes its inherited and global masks. Membership in Q_e , together with the tolerance inclusions, supplies usefulness and compatibility with its own triple. For every G_e -orbit $\mathcal{O}$, Lemma 4.1 gives

$$d_{e, \mathcal{O}} = \frac{|\{v \in \mathcal{O} : \neg P_e(v)\}|}{|\mathcal{O}|} \leq C_0/N.$$

For a passing variable v , use the event $R_{e, v}$ defined at the end of Section 3: it includes failure of e at the X uniqueness test and all compatible-target collisions relevant to v . Equation (26) gives

$$\Pr(R_{e, v} \mid S_e) \leq e^{-cN} \tag{28}$$

for one fixed $c > 0$, uniformly in e and v , for all large N . Here $P_e(v)$ is fixed before choosing the hash; the conditioning is still only S_e . We now average this individual loss over each full ideal orbit, including in the denominator the variables that fail the deterministic masks.

Specifically, put

$$r_{e,\mathcal{O}} = \frac{1}{|\mathcal{O}|} \sum_{\substack{v \in \mathcal{O} \\ P_e(v)}} \mathbf{1}_{R_{e,v}}.$$

By linearity of expectation, $\mathbb{E}(r_{e,\mathcal{O}} \mid S_e) \leq e^{-cN}$. Variables failing P_e contribute to $d_{e,\mathcal{O}}$, and need no collision estimate. If $P(N)$ bounds the total number of orbits on all three sides, Markov's inequality and a union bound imply

$$\Pr(\exists \mathcal{O} : r_{e,\mathcal{O}} > 1/N \mid S_e) \leq NP(N)e^{-cN} = e^{-cN+O(\log N)}.$$

Declare e good if it survives hashing, remains eligible, and all these random fractions are at most $1/N$. Including the exponentially small eligibility failure explicitly changes only the polynomial factor in this bound. Every good e therefore has missing fraction at most

$$\gamma_N = (C_0 + 1)/N \tag{29}$$

in each orbit. The eligible list remains the fixed list used in Section 3. Deterministic masks are accounted for as additional deletion flags; they do not trigger a recomputation of that list or of the compatibility candidates.

There are $\exp(NA_{\text{amb}} + O(\log N))$ target triples, and $\log M = N(A_{\text{amb}} - E + \delta) + O(1)$ with $\log |U| = \log M - o(N)$. If $\mathcal{G}$ is the set of good triples,

$$\begin{aligned} \mathbb{E}|\mathcal{G}| &\geq \exp(NA_{\text{amb}} + O(\log N)) h_M(1 - e^{-cN+O(\log N)}) \\ &= \exp(N(E - \delta) - o(N)). \end{aligned}$$

Thus one hash choice has at least this many good triples, up to integer rounding. This uses no independence between different targets or between their losses. The assignment proof has already made their holed outputs H_e disjoint. Discard all other outputs.

All strand lengths and alphabets are bounded, and there are $O(N)$ factors, so every ideal Q_e has at most $\exp(bN)$ monomials for one fixed b . For example, a product on at most aN original CW_5 sites has at most 18^{aN} terms before any masks, since CW_5 has $3 \cdot 5 + 3 = 18$ terms; any $b > a \log 18$ suffices. Apply Lemma 4.2 with $C = C_0 + 1$ to every good output. A common $R_N = \exp(O(N/\log N))$ works for all of them. Indeed,

$$\bigoplus_{r=1}^{R_N} \bigoplus_{e \in \mathcal{G}} H_e \cong \bigoplus_{e \in \mathcal{G}} \bigoplus_{r=1}^{R_N} H_e.$$

The same replication slots supply each summand with R_N copies. Within a summand choose its own covering family and rectangular maps, and keep variables of different summands distinct. This produces $\bigoplus_{e \in \mathcal{G}} Q_e$ with R_N copies of the original joint input. The number of summands does not multiply the replication cost.

Finally, exact target types give the same number of factors in every population, history, split-triple, and half class for every e . After repair, simultaneous position relabelings within each population put these classes in a fixed canonical order. They identify all Q_e with a single ideal product Q_N , without permuting the physical tensor sides. This canonicalization concerns the repaired ideal tensors and their child windows; no invariance of an earlier hole pattern is needed. Choose $0 < \delta < \min(E, \varepsilon)$ and then N large enough to absorb the $o(N)$ term. The result is at least $\exp(N(E - \varepsilon))$ identical ideal outputs from $\exp(o(N))$ input copies, as claimed. $\square$

4.4 Composition and subdivision

To iterate Theorem 3.1, we record the two algebraic operations explicitly. Write $D_r(T) = \bigoplus_{i=1}^r T$, and write $A \preceq B$ when A is a linear restriction of B .

Lemma 4.3 (Replication and composition). *Suppose $D_{s_1}(T) \preceq D_{r_1}(S)$ and $D_{s_2}(V) \preceq D_{r_2}(T)$. Then*

$$D_{s_1 s_2}(V) \preceq D_{r_1 r_2}(S).$$

For any finite collection of tensors,

$$\bigotimes_{i=1}^j D_{r_i}(F_i) \cong D_{\prod_{i=1}^j r_i} \left(\bigotimes_{i=1}^j F_i \right).$$

The same assertions permit different maps in different direct-sum branches, provided a common upper bound on their replication counts is used. Consequently every fixed finite sequence of the joint extractions above, including finite tensor products of them, costs only $\exp(o(N))$ replications of its original input.

Proof. Take r_2 copies of the first restriction and reorder their direct sum as $D_{s_1}(D_{r_2}(T))$. Apply the second restriction separately to its s_1 disjoint blocks. This gives the first identity with $r_1 r_2$ source copies, regardless of the value of s_1 . The second identity is direct-sum distributivity: its summands are indexed by the choices of one replication slot for each factor. A finished factor can be tensored through with replication count one. Branch-dependent maps act block by block after padding unused slots to the common bound. Lemma 4.2 provides such a bound uniformly over the selected outputs of each joint step. Products of finitely many $\exp(o(N))$ quantities are $\exp(o(N))$. For a fixed finite set of population specifications, all positive sizes are cN with $c > 0$ fixed, so a local $o(cN)$ is also $o(N)$. $\square$

There is also an exact way to subdivide a received population before its next use. Let B be its strand tensor, let ν_W be the prescribed statistic law on side W , and denote by $U(m, \nu, \eta)$ the restriction of $B^{\otimes m}$ to the three coordinatewise type windows with tolerance η . Partition the positions into fixed parts of sizes $m_1, \dots, m_j$, and on each part use the same laws and a positive tolerance $\eta_i \leq \eta$. There is an exact variable restriction

$$\bigotimes_{i=1}^j U(m_i, \nu, \eta_i) \preceq U \left(\sum_{i=1}^j m_i, \nu, \eta \right). \quad (30)$$

To see this, the empirical law of a concatenated variable is the weighted average of its partwise empirical laws, with error at most $\sum_i (m_i / \sum_j m_j) \eta_i \leq \eta$. Hence every retained variable satisfies the original window; after the partwise masks are imposed, no condition couples different parts, giving exactly the displayed product. Different side tolerances are handled in the same way. Apply this argument separately within every received history class; classes with different prescriptions remain distinct.

Equation (30) need not retain a large fraction of the earlier variables. The next application of Theorem 3.1 starts from this exact product, with its new positive population masses and its consistent inherited laws. Thus it supplies its own yield and mask analysis. Together with canonicalization after repair, this explains why arbitrary fixed finite subdivisions and heterogeneous populations can be used in a subsequent joint step. All such subdivisions and tolerances are fixed before N grows; no assertion of uniformity in a number of populations growing with N is needed.

The required tolerance hierarchy exists for any fixed finite calculation. At a step first make its global and compatibility tolerances small enough for the entropy errors allowed by its hash slack. Then choose its child windows smaller than all those tests and all inherited windows that they must satisfy. Repeat toward the descendants. There are finitely many constraints, each allowing a strictly positive choice. All constants in this section may depend on this fixed calculation; the strand limit is taken only after these choices.

5 Staggering the three extraction stages

A *lot* consists of N strands of length 8, hence of $8N$ copies of CW_5 . Its positive shapes pass successively through three stages: A splits length 8 into length 4, B splits length 4 into length 2, and C splits length 2 into length 1. Every shape with a zero coordinate terminates immediately as a matrix tensor. Theorem 3.1 lets one extraction act on populations from different stages. We specify their data first, then give the finite schedule that exploits this freedom.

5.1 Shapes, prescriptions, and the three stages

All coordinate labels in this section belong to $\{0, 1, 2\}$. A shape is *positive* if all its coordinates are positive. Choose a symmetric product-weight law on the ordered shapes of size 16: its probability at (a, b, c) is proportional to $\alpha_a \alpha_b \alpha_c$, where every $\alpha_j > 0$ and $a + b + c = 16$. Write A_g for the resulting probability of the sorted shape g . Thus the multiplicity of its distinct permutations is included in A_g . If ρ_g gives mass $1/3$ to each coordinate of g , counting multiplicities, the common marginal and initial ordinary yield are

$$\mu = \sum_g A_g \rho_g, \quad H_0 = H(\mu). \quad (31)$$

The product-weight hypothesis verifies maximum entropy at the given marginals. The initial ordinary extraction therefore yields $\exp(N(H_0 - o(1)))$ identical products of shape tensors, after the arbitrarily small losses of Theorem 3.1.

Give each sorted g canonical labels 0, 1, 2. Its six placements of these labels on the three physical sides will each have amount $NA_g/6$. To implement this when g has ties, let s_g be the number of label permutations fixing g . There are $6/s_g$ distinct physical shapes, each with $NA_g s_g/6$ positions. Divide its positions into s_g equal classes and assign the distinct tied-label placements to them. This is a regrouping of full tensor factors: no fine mask has yet been imposed. For example, each physical arrangement of $(4, 6, 6)$ has two such classes. Later laws may distinguish its two labels of weight 6. Descendants retain these canonical labels and are never re-sorted.

For each positive sorted g of size 16, choose a positive product-weight law $P_g(t)$ on

$$|t| = 8, \quad 0 \leq t \leq g,$$

with $P_g(t) = P_g(g-t)$. For every positive *ordered* t of size 8, likewise choose a positive product-weight law $p_t(u)$ on $|u| = 4$, $0 \leq u \leq t$, with $p_t(u) = p_t(t-u)$. The numerical data in Section 7 specify these laws, as well as the parameters $d_{t,u} \in [0, 1]$ and terminal laws z_t used below. Infeasible subscripts always have probability zero.

The length-2 statistic on one side is its unordered pair of weights. We use the following fixed coding; D_i counts the actual index words with statistic i , including the $q = 5$ choices at a weight-1

site:

i	0	1	2	3	4	5	$\kappa = (5, 4, 2, 3, 1, 0).$
weight pair	00	01	11	02	12	22	
r_i	0	1	2	2	3	4	
D_i	1	10	25	2	10	1	

(32)

Here r_i is the sum of the two weights, and κ is the image list for complementation. For positive t and a feasible child u , define the law $v_{t,u,W}$ on these six statistics, for each canonical label W , by requiring support on $r_i = u_W$. If $u_W \neq 2$ this determines the law. If $u_W = 2$, put masses $1 - d_{t,u}$ and $d_{t,u}$ on 11 and 02, respectively. A parameter is needed precisely when u is a permutation of 022 or 112.

On a length-4 strand the statistic is the *ordered* pair of the statistics of its length-2 halves. For positive t prescribe

$$V_{t,W} = \sum_u p_t(u) v_{t,u,W} \otimes v_{t,t-u,W}. \quad (33)$$

For a zero-coordinate t , let W_* be its first canonical label attaining $\max t$. Choose z_t on the ordered pairs (i, j) with $r_i + r_j = \max t$. Prescribe $V_{t,W_*} = z_t$, prescribe the point mass at $(0, 0)$ on zero coordinates, and prescribe the coordinatewise κ -image of z_t on the remaining nonzero coordinate, if any. The laws of the two matched sides therefore agree under complementation. The same property holds for zero-coordinate u : 01 complements 12, while 11 and 02 complement themselves. These are exactly the zero-side consistency requirements of Theorem 3.1.

Stage A is a sharing split of $T_{8,g}$ with coarse law P_g , child laws $V_{t,W}$, and canonical priority $(0, 1, 2)$. *Stage B* is a sharing split of $T_{4,t}$ with coarse law p_t , child laws $v_{t,u,W}$, and canonical priority $\pi(t)$. This priority is the lexicographically first permutation of the labels that reads t as one of

$$(1, 1, 6), \quad (1, 2, 5), \quad (1, 3, 4), \quad (2, 2, 4), \quad (3, 2, 3). \quad (34)$$

These represent every positive size-8 shape up to permutation. The last representative intentionally gives the middle priority to the coordinate of weight 2: for $t = (2, 3, 3)$ the priority is $(1, 0, 2)$. The symbols here remain inherited labels.

Stage C is an ordinary split of a positive $T_{2,u}$. Such u is a permutation of 112. Write its labels as (a, b, c) , where c is the distinguished label of weight 2. In this order, the four possible left shapes and their probabilities are

left shape	(1, 1, 0)	(0, 0, 2)	(1, 0, 1)	(0, 1, 1)
probability	$d_{t,u}/2$	$d_{t,u}/2$	$(1 - d_{t,u})/2$	$(1 - d_{t,u})/2$

(35)

The right shape is u minus the left one. The distinguished marginal fixes the first two probabilities, and the two Bernoulli-1/2 marginals fix the other two. Thus the coupling is uniquely determined by its marginals, also at $d = 0$ or 1, and has the required maximum entropy. Put

$$h_b(d) = H(1 - d, d) + d \log 2.$$

Its three ordinary capacities are $\log 2, \log 2, h_b(d_{t,u})$, with the last at label c . Every delivered length-1 shape has a zero coordinate and is already a matrix tensor.

This also verifies the inherited masks. At A to B, the input law required by B is exactly the mixture (33) prescribed by A; the orbit-mask estimate of Lemma 4.1 applies with smaller child tolerances. At B to C, an exact type in (35) forces unordered frequency $d_{t,u}$ on 02 and $1 - d_{t,u}$ on 11 at the distinguished label, and forces 01 at the other labels. Hence it satisfies the received $v_{t,u,W}$ masks exactly. Complementary children may have different d parameters: each is a separate factor, and both actual laws appear in (33).

After Stage B is repaired, its ideal children carry the prescribed $v_{t,u,W}$ windows. The earlier $V_{t,W}$ masks were constraints on the input to B; their holes have been repaired, so they are not additional constraints on the recovered children. Thus the exact Stage C types need only satisfy the $v_{t,u,W}$ windows, as verified above. The permutation group used for repair preserves the ideal child tensor, without needing to preserve the historical pattern of input holes.

A history records the lot, initial g , its label placement, every coarse split, its left/right provenance, and any subsequent part. We keep all these populations separate, even when their present shapes coincide. In particular equal u from different t never merge their $d_{t,u}$ values. The sums below record total amounts; they do not combine differently prescribed populations.

5.2 Capacity balance

Stages A and B have fixed capacities in their chosen priority orders. At Stage C, each positive length-2 population has two capacities equal to $\log 2$ and one equal to $h_b(d_{t,u})$. We may assign the latter to any priority position by choosing the extraction order. We use this freedom to make the sum of the three stages' capacity vectors have equal coordinates.

Amounts throughout are measured per initial length-8 strand. After summing over histories while retaining their specifications, the amounts of the smaller shapes are

$$m_t = 2 \sum_{\min g > 0} A_g P_g(t), \quad m_{t,u} = 2m_t p_t(u) \quad (\min t > 0). \quad (36)$$

The factors 2 count the left and right children; reflection symmetry of the coarse laws justifies each equality. A parent split is charged only once when computing its capacity.

Let $L_A \in \mathbb{R}^3$ be the sharing-capacity vectors of Stage A, weighted by A_g and summed over positive g , with entries in priority order. Likewise let L_B be the Stage B vectors weighted by m_t and read in the order $\pi(t)$. More explicitly, for each parent its vector is

$$(H(p_X), H(D_Y) - 2J_Y, H(D_Z) - 2J_Z),$$

using its own coarse law and child prescriptions in the definitions of Section 3. Thus the two vectors are fully determined by the data just specified. Let T be the total positive length-2 amount, and let B_* be the sum of those populations' average capacities, weighted by their amounts. The average over priority positions of the total stage capacity is C_* , where

$$\begin{aligned} T &= \sum_{\min t > 0, \min u > 0} m_{t,u}, & B_* &= \sum_{\min t > 0, \min u > 0} m_{t,u} \frac{2 \log 2 + h_b(d_{t,u})}{3}, \\ C_* &= B_* + \frac{1}{3} \sum_{i=0}^2 (L_A + L_B)_i. \end{aligned} \quad (37)$$

Suppose, as the numerical certificate verifies, that

$$T \log 2 > B_*, \quad C_* - (L_A + L_B)_i < T \log 2 \quad (0 \leq i < 3). \quad (38)$$

The denominator and each numerator below are positive by (38). The numbers

$$\lambda_i = \frac{T \log 2 - C_* + (L_A + L_B)_i}{3(T \log 2 - B_*)} \quad (0 \leq i < 3) \quad (39)$$

sum to 1: their numerators sum to $3(T \log 2 - B_*)$ by (37). Divide *each* positive length-2 history class into these proportions. In part i , place its distinguished label in priority position i and order

the other labels increasingly in the remaining positions. This changes the extraction order only; it keeps the same parent-specific $d_{t,u}$ and masks. The resulting Stage C capacity vector satisfies

$$\begin{aligned}(L_C)_i &= T \log 2 - \lambda_i \sum_{\min t > 0, \min u > 0} m_{t,u} (\log 2 - h_b(d_{t,u})) \\ &= T \log 2 - 3\lambda_i (T \log 2 - B_*) = C_* - (L_A + L_B)_i.\end{aligned}\tag{40}$$

Consequently $L_A + L_B + L_C = (C_*, C_*, C_*)$.

5.3 Physical orders and the finite schedule

Apply the subdivision restriction (30) separately within each received history class, keeping the same prescribed laws and using smaller positive windows. Different parts may use different widths. The resulting tensor is exactly the product of the partwise restrictions; no retained-fraction estimate is required, since Theorem 3.1 acts on this specified product directly. After each repair, exact coarse counts let simultaneous position permutations put its ideal child classes in a canonical order. All selected summands therefore have identical future inputs. These permutations are made after repair and need not preserve the earlier hole patterns.

Suppress the physical placement in a fixed canonical history h . At each of its six placements ϕ , its amount is $a_h N/6$ for the same a_h . This is true initially by the tied-label subdivision. Every child split and part allocation multiplies it by a number depending only on the canonical history, so it remains true inductively. If ρ_h maps priority positions to canonical labels, the physical priority is $\phi \circ \rho_h$. For any prescribed physical order σ there is exactly one contributing placement,

$$\phi = \sigma \circ \rho_h^{-1}.\tag{41}$$

Thus the six physical orders give a disjoint partition of the work, with one sixth of every history's priority-capacity vector in each factor. This applies to the Stage C parts as well as to A and B.

Assume that L_A, L_B, L_C have positive entries, as verified in Proposition 7.1. Then each vector obtained by summing any nonempty subset of the active stages also has positive minimum capacity. This supplies the condition $E > 0$ of Theorem 3.1 for each physical-order factor at every tick of the following schedule. Its finite applications will use the rational choices made in Section 6.1.

Fix an integer $K \geq 2$ and perform the initial ordinary extraction on each of K lots. Within every resulting summand, launch lot j at A on tick j , at B on tick $j + 1$, and at C on tick $j + 2$. The active stage patterns are

$$A; \quad A + B; \quad \underbrace{A + B + C; \dots; A + B + C}_{K-2 \text{ ticks}}; \quad B + C; \quad C.\tag{42}$$

Every lot is processed once at each of its three depths. We first compute the rates for the prescribed data. The rational approximations and arbitrarily small extraction losses are handled in Section 6.1.

At a tick, collect all currently active populations with one physical priority into one tensor factor, using (30) if a received type class needs subdivision. Apply one joint step to each of the six factors. A full tick has, in each factor, total priority-capacity vector

$$\frac{1}{6}(L_A + L_B + L_C) = \frac{1}{6}(C_*, C_*, C_*).$$

Its minimum is $C_*/6$, so multiplying the six output counts gives log-yield approaching NC_* . The four boundary ticks similarly have log-yields divided by N approaching

$$\min L_A, \quad \min(L_A + L_B), \quad \min(L_B + L_C), \quad \min L_C,$$

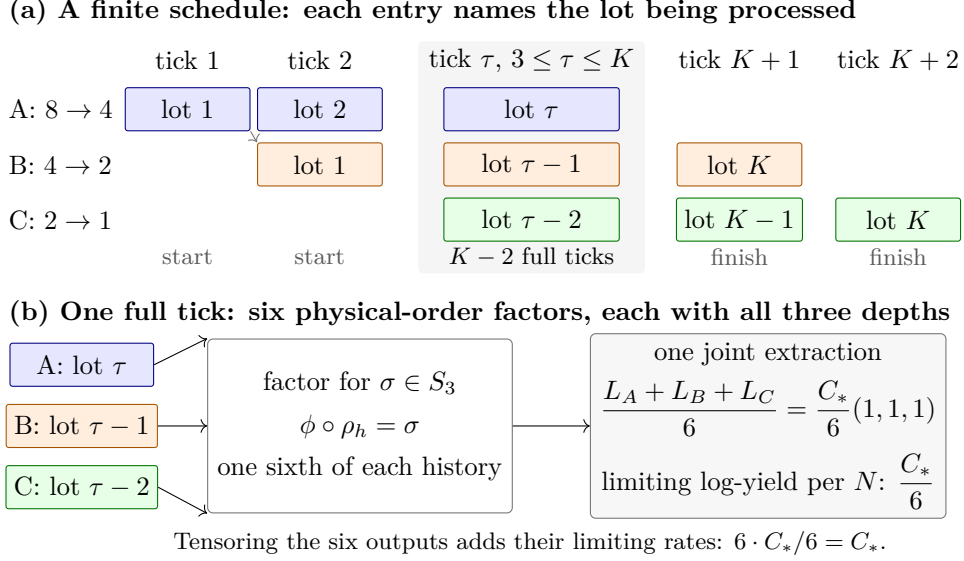


Figure 1: Mixed-depth extraction with finite boundaries ($K \geq 2$). A lot advances one stage per tick, while a full tick combines three different lots. Within each physical-order factor, the common order σ permits one application of Theorem 3.1 to the summed capacities. The four boundary ticks have the positive minima appearing in (43); they incur only the fixed deficit β . The middle column is absent when $K = 2$.

where $\min x = \min_{0 \leq i < 3} x_i$ for a vector x . All four minima are strictly positive by the entrywise positivity assumption. Relative to K full-tick contributions, their deficit is

$$\beta = 2C_* - \min L_A - \min(L_A + L_B) - \min(L_B + L_C) - \min L_C. \quad (43)$$

Using $L_A + L_B + L_C = C_* \mathbf{1}$, this is equivalently $\beta = (\max L_A - \min L_A) + (\max L_C - \min L_C) \geq 0$. The certificate in Proposition 7.1 also verifies $\beta < 1$. Thus the $K - 2$ full ticks and four boundary ticks contribute $KC_* - \beta$, and the initial extractions add KH_0 .

Figure 1 shows the distinction between progression of one lot and joint extraction of three distinct lots.

Finished matrix factors are carried through subsequent operations by tensor product. After the last tick, every lot has terminated, and exact coarse counts fix the number of terminal factors in every oriented history class. The next section computes their dimensions and then combines that count with the yield $K(H_0 + C_*) - \beta$ to obtain the rank constraint.

6 Terminal dimensions and the rank comparison

The terminal factors have either a zero coordinate or length one. Their dimensions can be counted directly, including the restrictions on their prescribed statistics. We first show that the completed ideal outputs restrict to one common oriented matrix tensor. We then combine its volume with the finite schedule's yield and compare ranks. Write

$$c_\ell(j) = [x^j](1 + 5x + x^2)^\ell$$

for the number of labeled weight words of length ℓ and weight sum j . With the amounts and laws of Section 5, define

$$S_0 = \sum_{\min g=0} A_g \log c_8(\max g), \quad (44)$$

$$S_1 = \sum_{\min t=0} m_t \left(H(z_t) + \sum_{i,j} z_t(i,j) \log(D_i D_j) \right), \quad (45)$$

$$S_2 = \sum_{\min t>0} \sum_{u \leq t, \sum_i u_i=4} m_{t,u} R_{t,u}, \quad (46)$$

$$S_* = S_0 + S_1 + S_2,$$

where g is sorted, t and u retain their coordinate labels, and

$$R_{t,u} = \begin{cases} 0, & \max u = 4, \\ \log 10, & \max u = 3, \\ h_b(d_{t,u}) + 2(1 - d_{t,u}) \log 5, & \max u = 2, \min u = 0, \\ (2 - d_{t,u}) \log 5, & \min u > 0. \end{cases} \quad (47)$$

In these sums, infeasible or zero-mass populations are omitted.

Proposition 6.1 (Leaf volumes and common dimensions). *Fix a finite number K of lots and rational stage laws satisfying the prescriptions of Section 5. Use fixed rational subdivisions retaining each history class's prescribed laws, as in (30); exact capacity balance is not required for this volume count. For all sufficiently divisible N , every completed ideal output in the staggered construction has a restriction to the same oriented matrix tensor $\langle a_N, b_N, c_N \rangle$, with*

$$\log(a_N b_N c_N) \geq K N S_* - O_K(\log(N + 2)). \quad (48)$$

The implied constant may depend on the fixed laws and subdivisions. For limiting real data, the corresponding volume rate can approach S_ arbitrarily closely by consistent rational approximation.*

Proof. We first count one terminal population. When a shape has a zero on side X , that side has a unique weight word and a unique variable. Every variable on Y has exactly one partner on Z : complement each weight $0 \leftrightarrow 2$ and retain the label at each weight 1. Consequently a set of M such matched pairs gives $\langle 1, 1, M \rangle$. A zero on Y gives $\langle M, 1, 1 \rangle$, and a zero on Z gives $\langle 1, M, 1 \rangle$. These orientations follow from the convention $x_{ij} y_{jk} z_{ki}$; in every case the volume is M . If two coordinates are zero, $M = 1$.

For an unrestricted length-eight leaf, the generating function above counts the choices on an axis of maximum weight. Complementation is a bijection to the other axis, so the volume is $c_8(\max g)$. Their total contribution per initial strand is therefore (44).

Here is the exact count needed for the restricted leaves. Suppose the statistic on the chosen nonzero side has values σ with multiplicities b_σ , and a population of m strands is restricted to an exact statistic type θ with integer counts $n_\sigma = m\theta_\sigma$. The number of surviving matching words is

$$M(m, \theta) = \frac{m!}{\prod_\sigma n_\sigma!} \prod_\sigma b_\sigma^{n_\sigma}. \quad (49)$$

Indeed, first choose the positions of each statistic and then its independent labeled realization at each position. The other side is uniquely determined. Prescribing the pushforward type under

complementation on that side retains all these pairs. For a fixed alphabet the multinomial estimate gives

$$\log M(m, \theta) = m \left(H(\theta) + \sum_{\sigma} \theta_{\sigma} \log b_{\sigma} \right) + O(\log(m+2)). \quad (50)$$

Zero type entries cause no difficulty, with $0! = 1$.

At a zero-coordinate length-four leaf, the statistic is the ordered pair (i, j) of the two length-two statistics. Its multiplicity is $D_i D_j$, and its type is z_t . Equation (50) therefore gives precisely the summand in (45). Within a length-two statistic, 11 has $5^2 = 25$ labelings and 02 has two orders; these choices have already been included in D . The two halves remain ordered. Thus, when z_t is specified by weights on $i \leq j$, an off-diagonal weight occurs twice in its normalizing sum, once for each ordered pair. There is no further multiplicity factor beyond $D_i D_j$ in (49).

The complement involution κ satisfies $D_{\kappa(i)} = D_i$ and sends (i, j) to $(\kappa(i), \kappa(j))$. The prescription on the second nonzero axis is exactly this pushforward of z_t . The same matching count applies when the two nonzero coordinates tie for the maximum: choosing the first maximum only chooses which of the two matched words is counted. No invariance of z_t under complementation is needed.

The nonnegative shapes of size four have sorted forms $(0, 0, 4)$, $(0, 1, 3)$, $(0, 2, 2)$, and $(1, 1, 2)$. These are the four disjoint cases of (47). The first has volume one. The second has ten matched words, since the weight-three side has either order of weights 1, 2 and five choices for the weight-one label. For the third, write $d = d_{t,u}$. In m strands, impose the exact type with $m(1-d)$ statistics 11 and md statistics 02. Its matching count is

$$\binom{m}{md} 25^{m(1-d)} 2^{md}.$$

Its logarithmic rate is $H(1-d, d) + 2(1-d) \log 5 + d \log 2$, the third case of (47). Complementation fixes both of these unordered statistics, so the same exact type is allowed on both nonzero sides.

For the positive case, consider first $u = (1, 1, 2)$ on the physical sides (X, Y, Z) , so Z is distinguished, and again put $d = d_{t,u}$. The four Stage C choices are

left and right shapes	probability	product of the two matrix tensors
$(0, 0, 2), (1, 1, 0)$	$d/2$	$\langle 1, 5, 1 \rangle$
$(1, 1, 0), (0, 0, 2)$	$d/2$	$\langle 1, 5, 1 \rangle$
$(0, 1, 1), (1, 0, 1)$	$(1-d)/2$	$\langle 5, 1, 5 \rangle$
$(1, 0, 1), (0, 1, 1)$	$(1-d)/2$	$\langle 5, 1, 5 \rangle$.

Every selected exact coarse type therefore produces, from m strands, the oriented tensor

$$\langle 5^{m(1-d)}, 5^{md}, 5^{m(1-d)} \rangle. \quad (51)$$

Its volume is $5^{m(2-d)}$, proving the final case of (47). This is the dimension of one output with fixed coarse blocks; the choices of coarse blocks have already contributed to the extraction yield. Permuting the physical sides gives the other orientations; the volume remains the same. This formula includes $d = 0$ and $d = 1$ whenever they are allowed by the chosen laws.

Both children must be counted in the volume sum. For each positive parent t , their contribution is

$$m_t \sum_u p_t(u) (R_{t,u} + R_{t,t-u}) = 2m_t \sum_u p_t(u) R_{t,u} = \sum_u m_{t,u} R_{t,u}.$$

The equality changes variables $u \mapsto t-u$ in the second sum and uses only $p_t(u) = p_t(t-u)$. It does not identify $d_{t,u}$ with $d_{t,t-u}$. The analogous calculation with P_g gives the amounts m_t after Stage A. Thus (44)–(46) count every terminal factor once.

Finally we verify that these counts provide common oriented dimensions, as required by Lemma 2.1. Keep separate every finite class specified by its lot, full ancestry, left or right half, coordinate-label placement on the physical sides, and subdivision for a priority choice. Exact coarse types fix the number of factors in each such class in every output. A simultaneous relabeling of positions on all three sides takes each class to the same canonical set of positions. Its ideal tensor and its prescribed type windows are consequently the same in every output. The recovery argument restores these ideal tensors before the terminal restrictions are made.

Choose once, in each zero-coordinate class with a statistic prescription, the exact matching type used in (49); use the same choice in every output. For rational data and divisible N , this type equals the prescribed law and lies in every positive tolerance window. Each class then has the same integer matching dimension and the same zero physical side; unrestricted classes already have fixed matching dimensions. At positive length-two classes the exact Stage C counts give the same oriented dimensions in (51) or its prescribed side permutation. Multiplying these fixed dimensions over all classes gives one common triple (a_N, b_N, c_N) , with each dimension fixed separately.

The parts of each subdivision retain the same prescribed laws, so their leading logarithmic contributions add to that of the original class. Allocating these parts among extraction priorities therefore does not change S_* . There are finitely many classes for fixed K . Summing (50) over them contributes only $O_K(\log(N+2))$ to the logarithm, while the leading term is $KN(S_0 + S_1 + S_2)$. This proves (48). For real limiting laws, choose consistent rational laws first and then a divisible N . The entropy function is continuous on each finite probability simplex, so the resulting normalized volume rates tend to S_* . $\square$

6.1 Finite extraction and the rank comparison

We now combine the schedule of Section 5 with the terminal dimensions just established. The finite statement records both the boundary deficit and the order in which the data are fixed.

Proposition 6.2 (Finite staggered yield). *Assume the prescriptions of Section 5, the balance conditions (38), and positive entries of L_A, L_B, L_C . Let $S_* = S_0 + S_1 + S_2$ be the terminal rate defined above. For every fixed integer $K \geq 2$ and every $\epsilon > 0$, suitable fixed rational approximations, part allocations, and tolerances give, for all sufficiently large divisible N , a restriction of $\exp(o_K(N))$ copies of $\text{CW}_5^{\otimes 8KN}$ to s_N identical matrix tensors of volume V_N , where*

$$\frac{\log s_N}{N} \geq K(H_0 + C_*) - \beta - \epsilon, \quad \frac{\log V_N}{N} \geq KS_* - \epsilon. \quad (52)$$

Proof. Use the initial extractions and the $K+2$ -tick schedule (42). Their total limiting logarithmic yield, divided by N , is $K(H_0 + C_*) - \beta$: each of the $K-2$ full ticks contributes C_* and the four boundaries contribute the positive minima in (43). We show that fixed rational data and a finite composition of joint steps attain these rates to within the stated losses.

Fix K before making any asymptotic choices. Choose rational product weights for the initial, A, and B laws, preserving their symmetries, and rational d and z data. Define v and V from these same choices by their formulas; do not approximate an inherited mixture separately. Choose rational allocations approaching (39). For these approximations the full-tick capacities approach $C_*/6$, which is all the yield bound requires. By continuity, the approximations may also be chosen to keep every boundary minimum strictly positive. There are finitely many histories, and all positive history masses are now fixed rational multiples of N . Omit zero classes and make N divisible by every required denominator. Choose the finitely many tolerances successively smaller toward the leaves, so all inherited masks and subdivisions meet Theorem 3.1.

There are K initial operations and at most $6(K + 2)$ subsequent joint operations. Lemma 4.3 pulls their finitely many replication factors back to $\exp(o_K(N))$ copies of the original source. This cost is per operation, including all its already disjoint summands; it is not repeated multiplicatively for each summand. Entropy continuity makes the total approximation and tolerance loss less than ϵ , for this fixed K , before N tends to infinity. Integer rounding and polynomial type-class factors contribute $o_K(N)$.

Proposition 6.1 applies to these fixed rational laws and subdivisions: the schedule keeps all histories distinct, and repair restores their ideal windows before any terminal restriction. Its common exact terminal types and exact Stage C counts give one oriented matrix tensor in every final summand, with volume rate approaching KS_* . Thus the approximation and extraction losses can be chosen to satisfy both bounds in (52). $\square$

Lemma 2.2 bounds the rank of the replicated source by

$$\exp(o_K(N)) 7^{8KN} \binom{24KN + 2}{2}.$$

Apply Lemma 2.1 to the identical outputs of Proposition 6.2, divide its logarithm by KN , and first let divisible $N \rightarrow \infty$. For fixed K let the approximation and tolerance losses tend to zero. The result is

$$H_0 + C_* - \frac{\beta}{K} + \frac{w}{3} S_* \leq 8 \log 7.$$

Letting $K \rightarrow \infty$ last gives the promised constraint

$$\boxed{H_0 + C_* + \frac{w}{3} S_* \leq 8 \log 7.} \quad (53)$$

Only fixed finite schedules were used; no estimate uniform in a simultaneously growing K is required.

7 Exact parameters and numerical certification

We now evaluate the logarithmic yield rate $H_0 + C_*$ and terminal logarithmic volume rate S_* for one choice of distributions. The same calculation verifies positive entries of the stage-capacity vectors, the balance conditions, and $S_* > 0$, then bounds the exponent ratio obtained from (53). Five integer arrays in Appendix A, followed by rational operations, specify this choice; no optimization procedure is part of the certificate.

For an integer argument n , define the rational weight

$$F(n) = \left(\sum_{j=0}^{16} \frac{(n/640000)^j}{j!} \right)^{64}. \quad (54)$$

The initial weights, the two split-weight arrays, and the terminal weights use 223 distinct arguments of F , including the implicit zero, all between -120770 and 45052 . The binary-parameter array instead supplies the fractions $n/10^6$ and is not passed to F . For the 223 weight arguments, exact rational evaluation gives

$$2^{-23} < F(n) < 2^{23}. \quad (55)$$

The truncated series is positive directly as well: for nonnegative arguments all terms are nonnegative, and for negative arguments the even Taylor partial sum lies above the positive exponential. This observation concerns ordinary real arithmetic used to choose probabilities, and imposes no restriction on the field of the tensor construction.

Complete decoding rules. All shapes and pairs below are traversed in increasing lexicographic order, and coordinate labels and array indices start at zero. For a shape s of size 2ℓ , write

$$\mathcal{B}(s) = \{b \in \mathbb{Z}_{\geq 0}^3 : b_0 + b_1 + b_2 = \ell, b_i \leq s_i (0 \leq i < 3)\}.$$

The arrays are consumed as follows.

1. The 17 entries $a_0, \dots, a_{16}$ define the initial sorted law by

$$A_g = \frac{\text{mult}(g) \prod_{i=0}^2 F(a_{g_i})}{\sum_{h_0 \leq h_1 \leq h_2, \sum h_i = 16} \text{mult}(h) \prod_{i=0}^2 F(a_{h_i})}, \quad \text{mult}(g) = \frac{3!}{\prod_j (\#\{i : g_i = j\})!}.$$

Thus the law on ordered shapes, before coalescing permutations, is a product of three one-coordinate weights.

2. The 118 entries in the Stage A array are read over all positive sorted g of size 16. The 54 entries in the Stage B array are read over all positive *ordered* t of size 8. At either parent s , set $\ell = \frac{1}{2} \sum_i s_i$. For coordinate $i = 0, 1, 2$, with $k = s_i$, read one integer $n_{s,i}(j)$ for each

$$\max(0, k - \ell) \leq j < \lfloor k/2 \rfloor, \quad n_{s,i}(\lfloor k/2 \rfloor) = 0$$

(the final zero consumes no entry). For $b \in \mathcal{B}(s)$ put

$$w_s(b) = \prod_{i=0}^2 F(n_{s,i}(\min(b_i, s_i - b_i))), \quad q_s(b) = \frac{w_s(b)}{\sum_{c \in \mathcal{B}(s)} w_s(c)}.$$

Use $P_g = q_g$ at Stage A and $p_t = q_t$ at Stage B. Each is a positive product-weight law and satisfies $q_s(b) = q_s(s - b)$ exactly.

3. Read the 75 entries of the fourth array over positive ordered t of size 8, and then over $u \in \mathcal{B}(t)$ with $\max u = 2$. An entry n means $d_{t,u} = n/10^6$. These include both the zero-coordinate 022 shapes and the positive 112 shapes. The range is $0 \leq d_{t,u} \leq 0.110106$.
4. Read the 36 entries of the fifth array over all ordered t of size 8 with a zero coordinate. For each t , list the pairs

$$(i, j), \quad 0 \leq i \leq j \leq 5, \quad r_i + r_j = \max t.$$

Read an integer for each pair except the last, whose integer is zero. Assign its F -weight to both (i, j) and (j, i) and normalize over *ordered* pairs to obtain z_t . Hence an off-diagonal weight occurs twice in the denominator, while a diagonal weight occurs once. The multiplicities $D_i D_j$ count words at the leaf and are not included in this probability normalization.

There are 30 sorted initial shapes, 21 positive sorted initial shapes, 21 positive ordered size-8 shapes, and 24 size-8 shapes with a zero. All specified parents consume their entries, including parents whose eventual amount is zero. For example, $(6, 1, 1)$ is included in the Stage B and binary traversals. Skipping it, or a zero-amount parent in the z traversal, would change the meaning of the arrays. The exact types in this construction are rational, because (54) and every normalization are rational. Only the balance fractions, and limiting tolerance choices, require subsequent rational approximation.

The complement prescriptions in Section 5 can be checked without logarithms. On the six statistic slots, κ is an involution with $D_{\kappa(i)} = D_i$ and $r_{\kappa(i)} = 4 - r_i$. At a 022 child it fixes both

slots 2 and 3, so the common parameter on the two nonzero sides gives exactly matching laws. At zero-coordinate length-four shapes the other side receives the pushforward of z_t in both slots. These identities, including all zero-axis cases, are checked with exact integers and fractions by the archived verifier. Complementary *child shapes* need not have equal parameters: for instance

$$d_{(1,3,4),(0,2,2)} = \frac{92966}{10^6}, \quad d_{(1,3,4),(1,1,2)} = \frac{1204}{10^6}.$$

They retain their separate prescriptions throughout the contractions.

Finite contractions. Here is an explicit evaluation rule for the conditional entropies appearing in the capacities. For a finite nonnegative vector x , let

$$\mathcal{K}(x) = \left(\sum_i x_i \right) \log \left(\sum_i x_i \right) - \sum_i x_i \log x_i, \quad 0 \log 0 = 0.$$

Thus $\mathcal{K}(x) = MH(x/M)$ when $M = \sum_i x_i > 0$, and $\mathcal{K}(0) = 0$. Given a parent s , its split law q , a priority (X, Y, Z) , and half-laws $Q_{b,W}$, form the ordered pair law

$$D_W(i, j) = \sum_{b \in \mathcal{B}(s)} q(b) Q_{b,W}(i) Q_{s-b,W}(j).$$

For $W = Y$, let $\mathcal{I}_W = \{b : b_Z = 0\}$; for $W = Z$, let $\mathcal{I}_W = \{b : b_X b_Y = 0\}$. The grouped entropy is exactly

$$J_W = \sum_{b \in \mathcal{I}_W} q(b) \mathcal{K}(Q_{b,W}) + \sum_{k=0}^{s_W} \mathcal{K} \left(\sum_{\substack{b \notin \mathcal{I}_W \\ b_W = k}} q(b) Q_{b,W} \right). \quad (56)$$

The capacity vector is

$$\left(\mathcal{K} \left(\left(\sum_{b: b_X = k} q(b) \right)_k \right), \mathcal{K}(D_Y) - 2J_Y, \mathcal{K}(D_Z) - 2J_Z \right).$$

At A substitute $(s, q, Q) = (g, P_g, V)$ and weight by A_g ; at B substitute $(t, p_t, v_{t,\cdot})$ and weight by m_t , with the priorities already specified. This rule uses each parent once. Child amounts and leaf volumes use both halves as in Proposition 6.1. Along with the formulas for $H_0, B_*, T, S_0, S_1, S_2$, it specifies only finite additions, multiplications, divisions, and logarithms of rational numbers.

Partial capacity	position 0	position 1	position 2
A: $\min g = 1$	.010739192	.016912163	.016213531
A: $\min g = 2$	.081941050	.088150511	.082591626
A: $\min g = 3$	.292033029	.283922596	.257330639
A: $\min g = 4$	.547143900	.527202791	.487211274
A: $\min g = 5$	.247999127	.248080948	.237330883
B: 116	.002770751	.002770751	.000443544
B: 125	.080908872	.106275766	.087117877
B: 134	.386356324	.454421695	.323639129
B: 224	.395346750	.395353092	.248930076
B: 233	.611253765	.686627626	.634289551

Table 1: Partial sums defining L_A and L_B . A B row sums over all ordered parents with that sorted shape. Each displayed entry has absolute error less than $4.795 \cdot 10^{-10}$.

H_0	1.841147374	B_*	1.158066972	T	2.345955063
S_0	.031704071	S_1	.882362236	S_2	11.680704014

Table 2: The other scalar contractions, each with absolute error less than $3 \cdot 10^{-8}$.

Proposition 7.1 (Numerical certificate). *The exact parameters above satisfy every positivity and balance condition used in Proposition 6.2. All thirty partial capacity entries in Table 1 are positive. In particular, $S_* > 0$, $T \log 2 - B_* > 0$, all entries of L_A, L_B, L_C are positive, and the three balance fractions are strictly positive. Moreover,*

$$5.612983956059236756 < H_0 + C_* < 5.612983956059236757, \quad (57)$$

$$12.594770321594676362 < S_* < 12.594770321594676363, \quad (58)$$

$$2.371054886006745684 < \frac{3(8 \log 7 - H_0 - C_*)}{S_*} < 2.371054886006745685. \quad (59)$$

With $\Omega = 2.371056$, the strict comparison has the outward enclosure

$$0.000004676829725968 < H_0 + C_* + \frac{\Omega}{3} S_* - 8 \log 7 < 0.000004676829725969. \quad (60)$$

Proof. We give the elementary interval rule and its error accounting, so that the certificate does not depend on assuming decimal logarithms are accurate. For $1 \leq y \leq 2$ put

$$P(y) = 2 \sum_{j=0}^{30} \frac{1}{2j+1} \left(\frac{y-1}{y+1} \right)^{2j+1}, \quad \mathcal{L}(x) = eP(2) + P(2^{-e}x), \quad e = \lfloor \log_2 x \rfloor.$$

For rational $x > 0$, its exponent e is determined by exact comparisons with powers of two. Integrating the geometric series for $2/(1-z^2)$ gives

$$0 \leq \log y - P(y) \leq \tau := \frac{2 \cdot 3^{-63}}{63(1-1/9)} < 3.120348 \cdot 10^{-32}.$$

Thus $|\log x - \mathcal{L}(x)| \leq (|e| + 1)\tau$, including when $e < 0$. On each dyadic interval $\mathcal{L}$ is increasing; at a boundary its left limit is $eP(2) + P(2) = (e+1)P(2)$, the value from the next interval. It is therefore continuous and increasing on $(0, \infty)$. Applying P at outward endpoints, then enlarging by the indicated tail, encloses $\log x$ even if an input interval crosses a dyadic boundary.

For completeness, all exponents in this evaluation satisfy $|e| < 2048$. There are fewer than 256 ordered points in every product-law support. By (55), a nonzero product-law probability exceeds 2^{-146} , a z entry exceeds 2^{-54} , and a positive binary entry exceeds 2^{-20} . The initial marginal entries exceed 2^{-148} . Positive entries of a length-four law V exceed 2^{-186} ; multiplying one by a Stage A probability gives 2^{-332} , and multiplying two gives 2^{-518} . These bounds also cover sums within nonzero groups; all these masses are at most one. The integer leaf multiplicities and coefficients are less than $7^8 < 2^{23}$. Hence the actual logarithm arguments lie between 2^{-518} and 2^{23} , a stronger range than needed. One may uniformly use

$$\epsilon = 2048\tau < 6.391 \cdot 10^{-29}. \quad (61)$$

Here is an independent propagation bound for replacing every logarithm literally by $\mathcal{L}$, without presuming that the resulting rational version of $\mathcal{K}$ is homogeneous. For a vector of total mass M , the error in $\mathcal{K}$ is at most $2M\epsilon$. The total singleton and nonsingleton group masses in (56) sum to one, so its error is at most 2ϵ . A sharing capacity has errors at most $(2, 6, 6)\epsilon$ in its three positions. The A parent amounts sum to at most one, the B parent amounts to at most two, and the positive length-two amount T is at most four. Consequently the entrywise error in $L_A + L_B$ is at most $(6, 18, 18)\epsilon$, and its average has error at most 14ϵ . The error in $h_b(d)$ is at most 3ϵ , hence that in B_* at most $20\epsilon/3$. Including H_0 gives

$$|\text{error in } (H_0 + C_*)| \leq \frac{68}{3}\epsilon.$$

For S_0, S_1, S_2 the corresponding bounds are $\epsilon, 6\epsilon, 20\epsilon$, respectively. Therefore

$$|\text{error in } S_*| \leq 27\epsilon, \quad |\text{error in the comparison at } \Omega| \leq \left(\frac{68}{3} + 9\Omega + 8\right)\epsilon < 3.324 \cdot 10^{-27}.$$

The file `verification/scripts/verify_parameters.py` implements the stated finite contractions. It constructs every $F(n)$ as an exact fraction, then uses outward interval operations for all arithmetic, including each normalizing denominator. Shared occurrences of a denominator are retained as interval occurrences; no independence or cancellation is assumed to reduce error. Exact structural checks establish normalization by division by the positive total, complement symmetry, and statistic support before the numerical checks. The interval checks of zero differences are only consistency checks of these identities. The primary verification evaluates the rational series above and adds its tail; a second run uses directed native logarithms as a cross-check. Binary interval endpoints are converted to exact fractions and serialized by floor for lower bounds and ceiling for upper bounds, including negative endpoints. These operations give outward decimal bounds without a nearest-rounding assumption. At 80 decimal digits, every reported native-log interval has width below 10^{-76} ; with the rational-series tail included, every reported interval has width below $3 \cdot 10^{-29}$. These bounds include all normalization and subsequent arithmetic rounding.

In addition to (57)–(60), the resulting bounds include

$$0.468025165562828972 < T \log 2 - B_* < 0.468025165562828973.$$

The following entries enclose L_C , the numerators $b_i = T \log 2 - C_* + (L_A + L_B)_i$ of the balance fractions, and the fractions λ_i , respectively. Each listed number is a lower bound; adding 10^{-15} to it gives an upper bound.

	0	1	2
L_C	1.115343821954349	.962118642424118	1.396738451843254
b	.510748315682386	.663973495212618	.229353685793481
λ	.363761291246081	.472890166361139	.163348542392779

The fractions sum to one exactly by their definition, since the sum of their numerators is $3(T \log 2 - B_*)$. Positivity of L_A, L_B follows already from the thirty positive partial entries. The boundary deficit used in the finite-lot schedule also satisfies

$$0.533798153 < 2C_* - \min_i (L_A)_i - \min_i (L_A + L_B)_i - \min_i (L_B + L_C)_i - \min_i (L_C)_i < 0.533798155 < 1.$$

These bounds verify all stated feasibility conditions. The script checks every displayed enclosure and stores the full interval outputs with the packaged integer data.

The rounded entries in Tables 1 and 2 already certify $\omega < 2.371056$. The stronger bound in Theorem 1.1 uses the interval enclosure (59). For the rounded-table check, allow independent errors of $4 \cdot 10^{-8}$ in all 30 partial capacities, H_0 , B_* , and the three volume rows. The total uncertainty in the comparison is at most

$$(12 + \Omega) 4 \cdot 10^{-8} = 5.7484224 \cdot 10^{-7}.$$

Using the displayed centers and an outward bound on $8 \log 7$ leaves a positive margin greater than $4.1011 \cdot 10^{-6}$. Thus the strict inequality has ample room for both arithmetic enclosure and the arbitrarily small losses in the construction. $\square$

Proof of Theorem 1.1. Since $S_* > 0$, Proposition 7.1 and (53) give

$$\omega \leq w \leq \frac{3(8 \log 7 - H_0 - C_*)}{S_*} < 2.371054886006746 < 2.371054887. \quad \square$$

The integer tables and their full traversal are sufficient to reproduce this comparison. The verification package includes a readable assignment table and both interval outputs; verification requires Python 3.10 or later and `mpmath` 1.3.0, with no optimization software or external data.

A Exact integer parameter tables

These five arrays, with the traversals in Section 7, specify all rational distributions used in the construction. Every integer is exact; line breaks carry no meaning. Array indices start at zero.

Initial shape weights: 17 entries (a).

-55839	-36137	-17449	-1780	11561	22640	31188	36731	38593	35898
29051	18387	3823	-14569	-39776	-38598	-23708			

Stage A split weights: 118 entries (P).

-3070	295	-20546	-1514	-38211	-9654	-7530	-2765	-55230	-16412
-14246	-5912	-78106	-32809	-7009	-26324	-13519	-3592	-96458	-47439
-16217	-46871	-25058	-9643	-95643	-62269	-28725	-8576	-2535	-2550
-34711	-7207	-3270	-6376	-53221	-15352	-3477	-14828	-4187	-74924
-32439	-7231	-3578	-24599	-8700	-92989	-46655	-16926	-3517	-42976
-20283	-5132	-107049	-63201	-31013	-9730	-3705	-67421	-35684	-13735
-68829	-36752	-13993	-7658	-7586	-76746	-33496	-7329	-7447	-17336
-4721	-96770	-49297	-17874	-7143	-30044	-10107	-116558	-68325	-33467
-10424	-7476	-51341	-24711	-5530	-76548	-41988	-15608	-17266	-4357
-17106	-4343	-120770	-70820	-34294	-10580	-16819	-4315	-32943	-9992
-83616	-44202	-16154	-17098	-4371	-56609	-25990	-5564	-56356	-25900
-5561	-33052	-9918	-32951	-9906	-57255	-26128	-5560		

Stage B split weights: 54 entries (p).

-39594	-5561	-55254	-15178	-62331	-19672	-54917	-18539	-16455	-32184
-5716	-11688	-5527	-53009	-6669	-6669	-64938	-20286	-6616	-26336
-26429	-6696	-60888	-19760	-6690	-5724	-32216	-15829	-58477	-18987
-26339	-6619	-26393	-26340	-26359	-6620	-16239	-55176	-18587	-55107
-18571	-16272	-60868	-19755	-6696	-6692	-55133	-18580	-16247	-32128
-5727	-32138	-5725	-122						

Binary parameters: 75 entries (d).

63	97655	0	42132	92966	1204	42148	95604	41743	3057
109933	42092	3747	41991	1	100622	42166	103562	9039	42157
103577	42165	63582	110082	4584	42175	104091	38838	102383	104793
42126	11814	63338	42165	104864	3587	42166	110086	2207	94768
42166	104185	4735	38889	110065	42166	102372	104314	38956	4876
102395	42166	110068	2953	42166	96000	41863	95723	3031	63321
42040	42174	104662	11860	104902	42189	2967	96042	42047	109872
3986	42200	3834	110106	60257					

Terminal length-four weights: 36 entries (z).

-11037	-36170	-4779	23107	-12831	29263	45052	22289	-6216	23895
-14326	-40670	-9623	-34674	-12708	-36610	-2174	21564	-5587	22521
-9766	28961	43809	22299	-12216	28778	44147	22176	-7811	21870
-6053	23786	-166	94	-52	17				

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